



## SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)  
Accredited by NAAC with 'A' Grade, UG Programmes CE, CSE, ECE, EEE, IT & ME are Accredited by NBA  
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

| Regulation: R20                                                                          |                                              | II / IV - B.Tech. I - Semester |      |    |   |    |            |            |             |
|------------------------------------------------------------------------------------------|----------------------------------------------|--------------------------------|------|----|---|----|------------|------------|-------------|
| COMPUTER SCIENCE & BUSINESS SYSTEM                                                       |                                              |                                |      |    |   |    |            |            |             |
| SCHEME OF INSTRUCTION & EXAMINATION<br>(With effect from 2020-21 admitted Batch onwards) |                                              |                                |      |    |   |    |            |            |             |
| Course Code                                                                              | Course Name                                  | Category                       | Cr   | L  | T | P  | Int. Marks | Ext. Marks | Total Marks |
| B20 BS 2104                                                                              | Applied Mathematical Techniques              | BS                             | 3    | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2101                                                                              | Formal Language and Automata Theory          | PC                             | 3    | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2102                                                                              | Computer Organization & Architecture         | PC                             | 3    | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2103                                                                              | Data Base Management Systems                 | PC                             | 3    | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2104                                                                              | Object Oriented Programming though JAVA      | ES                             | 3    | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2105                                                                              | Data Base Management Systems Lab             | PC                             | 1.5  | 0  | 0 | 3  | 15         | 35         | 50          |
| B20 CB 2106                                                                              | Computer Organization & Architecture Lab     | PC                             | 1.5  | 0  | 0 | 3  | 15         | 35         | 50          |
| B20 CB 2107                                                                              | Object Oriented Programming through JAVA Lab | ES                             | 1.5  | 0  | 0 | 3  | 15         | 35         | 50          |
| #SOC-I                                                                                   | Skill Oriented Course-I                      | SOC                            | 2    | 0  | 0 | 4  | --         | 50         | 50          |
| B20 MC 2102                                                                              | Professional Ethics & Human Values           | MC                             | 0    | 2  | 0 | 0  | --         | --         | --          |
| TOTAL                                                                                    |                                              |                                | 21.5 | 17 | 0 | 13 | 195        | 505        | 700         |

| #SOC-I | Course Code | Name of the Course         |
|--------|-------------|----------------------------|
|        | B20 IT 2108 | Network Administration     |
|        | B20 IT 2109 | spread Sheet Data Analysis |

| Course Code                                                                  | Category                                                                                                                                                                                                                                                                                                                                                                                                                      | L | T  | P  | C | I.M | E.M. | Exam   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|------|--------|
| B20BS2104                                                                    | BS                                                                                                                                                                                                                                                                                                                                                                                                                            | 3 | -- | -- | 3 | 30  | 70   | 3 Hrs. |
| APPLIED MATHEMATICAL TECHNIQUES                                              |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| (For CSBS)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| Prerequisites: calculus of functions of one and two variables                |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| Course Objectives: Students are expected to learn                            |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| 1.                                                                           | How to expand a periodic function in a Fourier series.                                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |      |        |
| 2.                                                                           | How to find Fourier transform for a given function and evaluate some real definite integrals.                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |
| 3.                                                                           | Formation and solution of first order linear and non linear partial differential equations                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |      |        |
| 4.                                                                           | The concept of interpolation and its use for equally and unequally spaced data points. Numerical methods to solve algebraic and transcendental equations.                                                                                                                                                                                                                                                                     |   |    |    |   |     |      |        |
| 5.                                                                           | Methods for numerical evaluation of integrals and for solving a first order ODE(IVP)                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| S.No                                                                         | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                       |   |    |    |   |     |      | K L    |
| 1                                                                            | Determine Fourier series and half range series of functions.                                                                                                                                                                                                                                                                                                                                                                  |   |    |    |   |     |      | K3     |
| 2                                                                            | Determine Fourier transforms of non-periodic functions and also use them to evaluate integrals.                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      | K3     |
| 3                                                                            | Form partial differential equations and solve Lagrange linear equation and nonlinear first order PDEs.                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |      | K3     |
| 4                                                                            | Solve linear higher order homogeneous and non-homogeneous PDEs.                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      | K3     |
| 5                                                                            | Fit an interpolation formula and perform interpolation for equally spaced and unequally spaced data.Determine a real root of an algebraic or a transcendental equation of one variable.                                                                                                                                                                                                                                       |   |    |    |   |     |      | K3     |
| 6                                                                            | Evaluate numerically certain definite integrals and solve a first order ordinary differential equation (IVP) by Euler and RK methods.                                                                                                                                                                                                                                                                                         |   |    |    |   |     |      | K3     |
| SYLLABUS                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| UNIT-I<br>(10 Hrs)                                                           | Fourier series: Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series.                                                                                                                                                                                                                               |   |    |    |   |     |      |        |
| UNIT-II<br>(10 Hrs)                                                          | Fourier Transforms: Fourier integral theorem (without proof), Fourier sine and cosine integrals, Fourier transform and its inverse, Sine and cosine transforms, Properties, Parseval's identity, inverse transforms, Finite Fourier transforms.                                                                                                                                                                               |   |    |    |   |     |      |        |
| UNIT-III<br>(12 Hrs)                                                         | PDEs of first order: Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions – Solutions of first order linear (Lagrange) equation and nonlinear (standard types) equations.<br>Linear higher order PDEs:<br>Solutions of linear partial differential equations with constant coefficients – RHS term of the type $e^{ax+by}$ , $\sin(ax+by)$ , $\cos(ax+by)$ , $x^m y^n$ . |   |    |    |   |     |      |        |

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| <b>UNIT-IV<br/>(12 Hrs)</b> | <p><b>Interpolation:</b> Introduction, forward differences, backward differences and Central differences. Differences of a polynomial, Newton's forward and backward interpolation formulae. Interpolation with unequal intervals: Newton's divided difference and Lagrange interpolation formulae.</p> <p><b>Solution of Algebraic and Transcendental Equations:</b><br/>Introduction, Bisection method, Method of false position, Iteration method &amp; Newton-Raphson method.</p> |
| <b>UNIT-V<br/>(10Hrs)</b>   | <p><b>Numerical Integration and solution of Ordinary Differential equations:</b><br/>Trapezoidal rule, Simpson's <math>1/3^{\text{rd}}</math> rule. Solution of first order ordinary differential equations by Taylor's method, Picard's method, Euler's method, Modified Euler's method, Fourth order Runge - Kutta method.</p>                                                                                                                                                      |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.                          | B. S. Grewal, Higher Engineering Mathematics, 43 <sup>rd</sup> Edition, Khanna Publishers.                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1.                          | Erwin Kreyszig, Advanced Engineering Mathematics, 10 <sup>th</sup> Edition, Wiley-India.                                                                                                                                                                                                                                                                                                                                                                                              |
| 2.                          | B. V. Ramana, Higher Engineering Mathematics, 2007 Edition, Tata Mc Graw Hill Education.                                                                                                                                                                                                                                                                                                                                                                                              |
| 3.                          | Steven C. Chapra, Applied Numerical Methods with MATLAB for Engineering and Science, Tata Mc Graw Hill Education.                                                                                                                                                                                                                                                                                                                                                                     |
| 4.                          | M. K. Jain, S.R.K. Iyengar and R.K. Jain, Numerical Methods for Scientific and Engineering Computation, New Age International Publications.                                                                                                                                                                                                                                                                                                                                           |

| Course Code                                                                  | Category                                                                                                                                                                                                                                                                                                                                                                                                                                        | L | T  | P  | C | I.M | E.M. | Exam   |     |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|------|--------|-----|
| B20CB2101                                                                    | PC                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3 | -- | -- | 3 | 30  | 70   | 3 Hrs. |     |
| FORMAL LANGUAGE & AUTOMATA THEORY                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |     |
| (For CSBS)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |     |
| Course Objectives: Students are expected to learn                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |     |
| 1.                                                                           | To learn fundamentals of Regular and Context Free Grammars and Languages                                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |      |        |     |
| 2.                                                                           | To understand the relation between Regular Language and Finite Automata and machines                                                                                                                                                                                                                                                                                                                                                            |   |    |    |   |     |      |        |     |
| 3.                                                                           | To learn how to design Automata's and machines as Acceptors, Verifiers and Translators                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |     |
| 4.                                                                           | To understand the relation between Contexts free Languages, PDA and TM                                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |     |
| 5.                                                                           | To learn how to design PDA as acceptor and TM as Calculators                                                                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |      |        |     |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |     |
| S.No                                                                         | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |    |    |   |     |      |        | K L |
| 1                                                                            | Ability to analyze and construct Finite Automata from a regular expression ,regular grammar or regular language.                                                                                                                                                                                                                                                                                                                                |   |    |    |   |     |      |        | K4  |
| 2                                                                            | Ability to analyze and construct a PDA from a given CFG or CFL                                                                                                                                                                                                                                                                                                                                                                                  |   |    |    |   |     |      |        | K4  |
| 3                                                                            | Ability to analyze and construct a Turing Machine from a REL or Unrestricted Grammar                                                                                                                                                                                                                                                                                                                                                            |   |    |    |   |     |      |        | K4  |
| SYLLABUS                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |    |    |   |     |      |        |     |
| UNIT-I<br>(10Hrs)                                                            | Finite Automata: Need of Automata theory, Central Concepts of Automata Theory, Automation, Finite Automation, Transition Systems, Acceptance of a String, DFA, Design of DFAs, NFA, Design of NFA, Equivalence of DFA and NFA, Conversion of NFA into DFA, Finite Automata with $\epsilon$ -Transitions, Minimization of Finite Automata, Finite Automata with output-Mealy and Moore Machines, Applications and Limitation of Finite Automata. |   |    |    |   |     |      |        |     |
| UNIT-II<br>(08Hrs)                                                           | Regular Expressions, Regular Sets, Identity Rules, Equivalence of two RE, Manipulations of REs, Finite Automata and Regular Expressions, Inter Conversion, Equivalence between FA and RE, Pumping Lemma of Regular Sets, Closure Properties of Regular Sets, Grammars, Classification of Grammars, Chomsky Hierarchy Theorem, Right and Left Linear Regular Grammars, Equivalence between RG and FA, Inter Conversion                           |   |    |    |   |     |      |        |     |
| UNIT-III<br>(10Hrs)                                                          | Formal Languages, Context Free Grammar, Leftmost and Rightmost Derivations, Parse Trees, Ambiguous Grammars, Simplification of Context Free Grammars-Elimination of Useless Symbols, $\epsilon$ -Productions and Unit Productions, Normal Forms-Chomsky Normal Form and Greibach Normal Form, Pumping Lemma, Closure Properties, Applications of Context Free Grammars.                                                                         |   |    |    |   |     |      |        |     |
| UNIT-IV<br>(10Hrs)                                                           | Pushdown Automata, Definition, Model, Graphical Notation, Instantaneous Description, Language Acceptance of Pushdown Automata, Design of Pushdown Automata,                                                                                                                                                                                                                                                                                     |   |    |    |   |     |      |        |     |

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|                             | Deterministic and Non – Deterministic Pushdown Automata, Equivalence of Pushdown Automata and Context Free Grammars, Conversion, Two Stack Pushdown Automata, Application of Pushdown Automata.                                                                                                                                                                                                      |
| <b>UNIT-V<br/>(12 Hrs.)</b> | <b>Turning Machine:</b> Definition, Model, Representation of TMs-Instantaneous Descriptions, Transition Tables and Transition Diagrams, Language of a TM, Design of TMs, Types of TMs, Church's Thesis, Universal and Restricted TM, Decidable and Undecidable Problems, Halting Problem of TMs, Post's Correspondence Problem, Modified PCP, Classes of P and NP, NP-Hard and NP-Complete Problems. |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.                          | Introduction to Automata Theory, Languages, and Computation John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman.                                                                                                                                                                                                                                                                                  |
| 2.                          | Theory of Computer Science-Automata, Languages and Computation, K. L. P. Mishra and N. Chandrasekharan, 3rd Edition, PHI, 2007                                                                                                                                                                                                                                                                       |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1.                          | Elements of the Theory of Computation, Harry R. Lewis and Christos H. Papadimitriou.                                                                                                                                                                                                                                                                                                                 |
| 2.                          | Theory of Computation, V. Kulkarni, Oxford University Press, 2013                                                                                                                                                                                                                                                                                                                                    |
| 3.                          | Theory of Automata, Languages and Computation, Rajendra Kumar, McGraw Hill, 2014                                                                                                                                                                                                                                                                                                                     |
| <b>e-Resources:</b>         |                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1                           | <a href="https://nptel.ac.in/courses/106/104/106104028/">https://nptel.ac.in/courses/106/104/106104028/</a>                                                                                                                                                                                                                                                                                          |

| Course Code                                                                  | Category                                                                                                                                                                                                                                                                                                                                                                                                                                                    | L | T  | P  | C | I.M | E.M. | Exam   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|------|--------|
| B20CB2102                                                                    | PC                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3 | -- | -- | 3 | 30  | 70   | 3 Hrs. |
| COMPUTER ORGANIZATION & ARCHITECHTURE                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      |        |
| (For CSBS)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      |        |
| Course Objectives: Students are expected to learn                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      |        |
| 1.                                                                           | Principles and the Implementation of ComputerArithmetic                                                                                                                                                                                                                                                                                                                                                                                                     |   |    |    |   |     |      |        |
| 2.                                                                           | Operation of CPUs including RTL, ALU, Instruction Cycle and Busses                                                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |
| 3.                                                                           | FundamentalsofdifferentInstructionSetArchitecturesandtheirrelationshipptothe CPU design                                                                                                                                                                                                                                                                                                                                                                     |   |    |    |   |     |      |        |
| 4.                                                                           | Memory System and I/O Organization                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |
| 5.                                                                           | Principles of Operation of Multiprocessor Systems and Pipelining                                                                                                                                                                                                                                                                                                                                                                                            |   |    |    |   |     |      |        |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      |        |
| S.No                                                                         | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |    |    |   |     |      | K L    |
| 1                                                                            | Illustrate the various data representations, notations, arithmetic algorithms and flow control for various instructions using micro operations in basic computer                                                                                                                                                                                                                                                                                            |   |    |    |   |     |      | K2     |
| 2                                                                            | Detailed understanding of architecture and functionality of central processing unit and various control units                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      | K2     |
| 3                                                                            | Exemplify in a better way the I/O and memory organization                                                                                                                                                                                                                                                                                                                                                                                                   |   |    |    |   |     |      | K3     |
| 4                                                                            | Illustrate concepts of parallel processing, pipelining and inter processor communication                                                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |      | K2     |
| SYLLABUS                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      |        |
| UNIT-I<br>(10Hrs)                                                            | Basic Structure of Computers: Basic Organization of Computers, Historical Perspective, Bus Structures, Data Representation: Data types, Complements, Fixed Point Representation. Floating, Point Representation. Other Binary Codes, Error Detection Codes.<br>Computer Arithmetic: Addition and Subtraction, Multiplication Algorithms, Division Algorithms.                                                                                               |   |    |    |   |     |      |        |
| UNIT-II<br>(08Hrs)                                                           | Register Transfer Language and Micro operations: Register Transfer language. Register Transfer Bus and Memory Transfers, Arithmetic Micro operations, Logic Micro Operations, Shift Micro Operations, Arithmetic Logic Shift Unit.<br>Basic Computer Organization and Design: Instruction Codes, Computer Register, Computer Instructions, Instruction Cycle, Memory – Reference Instructions. Input – Output and Interrupt, Complete Computer Description. |   |    |    |   |     |      |        |
| UNIT-III<br>(10Hrs)                                                          | Central Processing Unit: General Register Organization, STACK Organization. Instruction Formats, Addressing Modes, Data Transfer and Manipulation, Program Control, Reduced Instruction Set Computer.<br>Micro programmed Control: Control Memory, Address Sequencing, Micro Program example, Design of Control Unit.                                                                                                                                       |   |    |    |   |     |      |        |

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| <b>UNIT-IV<br/>(10Hrs)</b>  | <b>Memory Organization:</b> Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.<br><b>Input-Output Organization:</b> Peripheral Devices, Input-Output Interface, Asynchronous data transfer, Modes of Transfer, Priority Interrupts, Direct Memory Access. |
| <b>UNIT-V<br/>(12 Hrs.)</b> | <b>Multi Processors:</b> Introduction, Characteristics of Multiprocessors, Inter connection Structures, Inter Processor Arbitration.<br><b>Pipeline:</b> Parallel Processing, Pipelining, Instruction Pipeline.                                                                                           |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                           |
| 1.                          | Computer System Architecture M. M. Mano:, 3rd ed., Prentice Hall of India, New Delhi, 1993                                                                                                                                                                                                                |
| 2.                          | Computer Organization, Carl Hamacher, Zvonko Vranesic, Safwat Zaky, 5/e, McGraw Hill, 2002.                                                                                                                                                                                                               |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                           |
| 1.                          | Computer Organization and Architecture, William Stallings, 6/e, Pearson, 2006.                                                                                                                                                                                                                            |
| 2.                          | Structured Computer Organization, Andrew S. Tanenbaum, 4/e, Pearson, 2005.                                                                                                                                                                                                                                |
| 3.                          | Fundamentals of Computer Organization and Design, Sivarama P. Dandamudi, Springer, 2006.                                                                                                                                                                                                                  |
| <b>e-Resources</b>          |                                                                                                                                                                                                                                                                                                           |
| 1.                          | <a href="https://nptel.ac.in/courses/106/105/106105163/">https://nptel.ac.in/courses/106/105/106105163/</a>                                                                                                                                                                                               |
| 2.                          | <a href="http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf">http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf</a>                                                                                                                                                                                           |

| Code                                                                        | Category                                                                                                                                                                                                                                                                              | L | T  | P  | C | I.M | E.M | Exam   |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|-----|--------|
| B20CB2103                                                                   | PC                                                                                                                                                                                                                                                                                    | 3 | -- | -- | 3 | 30  | 70  | 3 Hrs. |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| DATA BASE MANAGEMENT SYSTEMS                                                |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| (For CSBS)                                                                  |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| Course Objectives: Students are expected to learn                           |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| 1                                                                           | To introduce about database management systems.                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| 2                                                                           | To give a good formal foundation on the E R Model and relational model of data.                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| 3                                                                           | To introduce the concepts of basic SQL as a universal Database language.                                                                                                                                                                                                              |   |    |    |   |     |     |        |
| 4                                                                           | To demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization.                                                                                                                                       |   |    |    |   |     |     |        |
| 5                                                                           | To provide an overview of physical design of a database system, by discussing Database indexing techniques and storage techniques.                                                                                                                                                    |   |    |    |   |     |     |        |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| Course Outcomes:After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| S.No                                                                        | Outcome                                                                                                                                                                                                                                                                               |   |    |    |   |     |     | K L    |
| 1.                                                                          | Understand the basic concepts of Database systems                                                                                                                                                                                                                                     |   |    |    |   |     |     | K2     |
| 2.                                                                          | Analyze requirements of an organization and develop a database schema in terms of E R model and Relational model.                                                                                                                                                                     |   |    |    |   |     |     | K4     |
| 3                                                                           | Analyze a query and formulate optimal solution using the knowledge of query languages like SQL.                                                                                                                                                                                       |   |    |    |   |     |     | K4     |
| 4                                                                           | Design well structured relations by applying normalization to remove anomalies in relations.                                                                                                                                                                                          |   |    |    |   |     |     | K4     |
| 5                                                                           | Analyze the issues in transaction execution, data storage and Security                                                                                                                                                                                                                |   |    |    |   |     |     | K2     |
| SYLLABUS                                                                    |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| UNIT-I<br>(9Hrs)                                                            | Introduction: Introduction to Database Systems, File system Vs DBMS, Brief introduction to Hierarchical, Network and Relational Models.<br>Database system architecture: Database structure, Data Abstraction, Data Independence, 2-tier,3-tier architectures, Schemas and Instances. |   |    |    |   |     |     |        |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| UNIT-II<br>(10Hrs)                                                          | Database Design:<br>E-R model: entities, relationships, cardinalities, participation constraints, weak entity sets, hierarchies and aggregation.<br>Relational Model: Introduction to SQL, DDL,DML and DCL, Integrity constraints, mapping ER diagrams to relations, views.           |   |    |    |   |     |     |        |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| UNIT-III<br>(12Hrs)                                                         | Query Processing: simple queries, sub queries, joins, Aggregate functions, group by and having clause, set operations.<br>Query Optimization: Evaluation of relational algebraic expressions, Query Equivalence, join strategies.                                                     |   |    |    |   |     |     |        |
|                                                                             |                                                                                                                                                                                                                                                                                       |   |    |    |   |     |     |        |
| UNIT-IV<br>(10Hrs)                                                          | Schema Refinement and Normalization: Purpose of Normalization, keys, functional dependencies, Normal forms: 1NF, 2NF, 3NF and BCNF, loss-less join, dependency preservation, MVDs fourth normal form, Join dependencies and fifth normal form.                                        |   |    |    |   |     |     |        |

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| <b>UNIT-V<br/>(10Hrs)</b> | <b>Transaction Processing:</b> Transaction concepts, <b>ACID</b> properties, schedules, Concurrent execution of transactions, concurrency control using 2PL.<br><b>Database Security:</b> Authentication, Authorization and access control, Intrusion detection, SQL injection.<br><b>Indexing:</b> Primary, Secondary, and Cluster indexes, Hash based indexing and B+-tree Indexing. |
|                           |                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Text Books:</b>        |                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.                        | Database System Concepts. Abraham Silberschatz, Henry F. Korth and S. Sudarshan.                                                                                                                                                                                                                                                                                                       |
|                           |                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reference Books:</b>   |                                                                                                                                                                                                                                                                                                                                                                                        |
| 1.                        | Principles of Database and Knowledge – Base Systems, Vol 1 by J. D. Ullman.                                                                                                                                                                                                                                                                                                            |
| 2                         | Fundamentals of Database Systems. R. Elmasri and S. Navathe.                                                                                                                                                                                                                                                                                                                           |
| 3                         | Foundations of Databases. Serge Abiteboul, Richard Hull, Victor Vianu.                                                                                                                                                                                                                                                                                                                 |

| Course Code                                                                  | Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | L | T  | P  | C | I.M | E.M | Exam   |
|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|-----|--------|
| B20CB2104                                                                    | ES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3 | -- | -- | 3 | 30  | 70  | 3 Hrs. |
| OBJECT ORIENTED PROGRAMMING THROUGH JAVA                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| (For CSBS)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| Course Objectives: Students are expected to learn                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| 1.                                                                           | To identify Java language components and how they work together in applications                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| 2.                                                                           | To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.                                                                                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |     |        |
| 3.                                                                           | To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling in Java applications                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| 4.                                                                           | To understand how to design applications with threads in Java                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |     |        |
| 5.                                                                           | To understand how to use Java APIs for program development                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |    |    |   |     |     |        |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| S. No                                                                        | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |    |    |   |     |     | K L    |
| 1                                                                            | Able to realize the concept of Object Oriented Programming & Java Programming Constructs                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |    |    |   |     |     | K3     |
| 2                                                                            | Able to describe the basic concepts of Java such as operators, classes, objects, inheritance, packages, Enumeration and various keywords                                                                                                                                                                                                                                                                                                                                                                                                  |   |    |    |   |     |     | K2     |
| 3                                                                            | Apply the concept of exception handling and Input/ Output operations                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |     | K3     |
| 4                                                                            | Able to design the applications of Java & Java applet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |    |    |   |     |     | K4     |
| 5                                                                            | Able to Analyze & Design the concept of Event Handling and Abstract Window Toolkit                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     | K4     |
| SYLLABUS                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |    |    |   |     |     |        |
| UNIT-I<br>(10Hrs)                                                            | Introduction to OOP: Introduction, Need of Object Oriented Programming, Principles of Object Oriented Languages, Procedural languages Vs OOP, Applications of OOP, History of JAVA, Java Virtual Machine, Java Features, Program structures.<br>Programming Constructs: Variables, Primitive Data types, Identifiers- Naming Coventions, Keywords, Literals, Operators-Binary, Unary and ternary, Expressions, Precedence rules and Associativity, Primitive Type Conversion and Casting, Flow of control- Branching, Conditional, loops. |   |    |    |   |     |     |        |
| UNIT-II<br>(08Hrs)                                                           | Classes and Objects- classes, Objects, Creating Objects, Methods, constructors- Constructor overloading, cleaning up unused objects-Garbage collector, Class variable and Methods-Static keyword, this keyword, Arrays, Command line arguments                                                                                                                                                                                                                                                                                            |   |    |    |   |     |     |        |
| UNIT-III<br>(10Hrs)                                                          | Inheritance: Types of Inheritance, Deriving classes using extends keyword, Method overloading, method overriding, dynamic method dispatch, super keyword, final keyword, Abstract class.<br>Interfaces, Packages and Enumeration: Interface-Extending interface, Interface Vs Abstract classes, Packages-Creating packages, using Packages, Access protection, java.                                                                                                                                                                      |   |    |    |   |     |     |        |

|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | lang package. Exceptions & Assertions – Introduction, Exception handling techniques- try... catch, throw, throws, finally block, user defined exception, Exception Encapsulation and Enrichment, Assertions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>UNIT-IV<br/>(10Hrs)</b>  | <p><b>Multi Threading:</b> java.lang.Thread, The main Thread, Creation of new threads, Thread priority, Multithreading- Using isAlive () and join (), Synchronization, suspending and Resuming threads, Communication between Threads.</p> <p><b>Input/Output:</b> reading and writing data, java.io package</p> <p>Stream based I/O (java.io) – The Stream classes-Byte streams and Character streams, Reading console Input and Writing Console Output, File class, Reading and writing Files.</p>                                                                                                                                                                                                                                                                                                                                           |
| <b>UNIT-V<br/>(12 Hrs.)</b> | <p><b>GUI Programming with Swing–Introduction,</b> limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout.</p> <p><b>Event Handling-</b> The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events, Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application,</p> <p><b>Applets</b> – Applets and HTML, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JCombo Box, Swing Menus, Dialogs.</p> |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.                          | Java one step ahead, Anitha seth, B.L.Juneja, oxford.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 2.                          | Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt. Ltd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 3.                          | Object Oriented Programming with Java, essentials and applications, Rajkumar Bhuyya,selvi, chu TMH.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.                          | Introduction to Java programming, a primer , Balaguruswamy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2.                          | Object Oriented Programming through Java, P. Radha Krishna                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

| Code                                                                             | Category                                                                                                                                                                                                                                                                                                                                                                    | L  | T  | P | C   | I.M | E.M | Exam   |    |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|--------|----|
| B20CB2105                                                                        | PC                                                                                                                                                                                                                                                                                                                                                                          | -- | -- | 3 | 1.5 | 15  | 35  | 3 Hrs. |    |
|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| DATA BASE MANAGEMENT SYSTEMS LAB                                                 |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| (For CSBS)                                                                       |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| Course Objectives: The student who successfully completes this course will have: |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 1                                                                                | Populate and query a database using SQL DDL/DML Commands                                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        |    |
| 2                                                                                | Declare and enforce integrity constraints on a database                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 3                                                                                | Writing Queries using advanced concepts of SQL                                                                                                                                                                                                                                                                                                                              |    |    |   |     |     |     |        |    |
| 4                                                                                | Programming PL/SQL including procedures, functions, cursors and triggers                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        |    |
|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| Course Outcomes: After completion of the course, the student will be able to     |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| S. No                                                                            | Outcome                                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        | KL |
| 1                                                                                | Utilize SQL to execute queries for creating database and performing data manipulation operations                                                                                                                                                                                                                                                                            |    |    |   |     |     |     |        | K2 |
| 2                                                                                | Examine integrity constraints to build efficient databases                                                                                                                                                                                                                                                                                                                  |    |    |   |     |     |     |        | K1 |
| 3                                                                                | Apply Queries using Advanced Concepts of SQL                                                                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        | K3 |
| 4                                                                                | Build PL/SQL programs including stored procedures, functions, cursors and triggers                                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |        | K2 |
|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| LIST OF EXPERIMENTS                                                              |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
|                                                                                  |                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 1                                                                                | Creation, altering and dropping of tables and inserting rows into a table (use constraints while creating tables) examples using SELECT command.                                                                                                                                                                                                                            |    |    |   |     |     |     |        |    |
| 2                                                                                | Queries (along with sub Queries) using ANY, ALL, IN, EXISTS, NOTEXISTS, UNION, INTERSET, Constraints. Example:- Select the roll number and name of the student who secured fourth rank in the class.                                                                                                                                                                        |    |    |   |     |     |     |        |    |
| 3                                                                                | Queries using Aggregate functions (COUNT, SUM, AVG, MAX and MIN), GROUP BY, HAVING and Creation and dropping of Views.                                                                                                                                                                                                                                                      |    |    |   |     |     |     |        |    |
| 4                                                                                | Queries using Conversion functions (to_char, to_number and to_date), string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc, round, to_char, to_date)                                                                 |    |    |   |     |     |     |        |    |
| 5                                                                                | Creation of simple PL/SQL program which includes declaration section, executable section and exception –Handling section (Ex. Student marks can be selected from the table and printed for those who secured first class and an exception can be raised if no records were found)<br>Insert data into student table and use COMMIT, ROLLBACK and SAVEPOINT in PL/SQL block. |    |    |   |     |     |     |        |    |
| 6                                                                                | Develop a program that includes the features NESTED IF, CASE and CASE expression. The program can be extended using the NULLIF and COALESCE functions.                                                                                                                                                                                                                      |    |    |   |     |     |     |        |    |
| 7                                                                                | Program development using WHILE LOOPS, numeric FOR LOOPS, nested loops using ERROR Handling, BUILT –IN Exceptions, USE defined Exceptions, RAISE-APPLICATION ERROR                                                                                                                                                                                                          |    |    |   |     |     |     |        |    |

|    |                                                                                                                          |
|----|--------------------------------------------------------------------------------------------------------------------------|
| 8  | Programs development using creation of procedures, passing parameters IN and OUT of 37 PROCEDURES                        |
| 9  | Program development using creation of stored functions, invoke functions in SQL Statements and write complex functions.  |
| 10 | Develop programs using features parameters in a CURSOR, FOR UPDATE CURSOR, WHERE CURRENT of clause and CURSOR variables. |
| 11 | Develop Programs using BEFORE and AFTER Triggers, Row and Statement Triggers and INSTEAD OF Triggers                     |
| 12 | Create a table and perform the search operation on table using indexing and non-indexing techniques.                     |

#### **Reference Books:**

|    |                                                                                    |
|----|------------------------------------------------------------------------------------|
| 1. | Oracle: The Complete Reference by Oracle Press                                     |
|    | Nilesh Shah, "Database Systems Using Oracle", PHI, 2007                            |
| 2. | Rick F Vander Lans, "Introduction to SQL", Fourth Edition, Pearson Education, 2007 |

| Code                                                                                      | Category                                                                                                                                                                                                                                                                       | L  | T  | P | C   | I.M | E.M | Exam   |    |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|--------|----|
| B20CB2106                                                                                 | PC                                                                                                                                                                                                                                                                             | -- | -- | 3 | 1.5 | 15  | 35  | 3 Hrs. |    |
| COMPUTER ORGANIZATION & ARCHITECTURE LAB                                                  |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| (For CSBS)                                                                                |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| Course Objectives: The student who successfully completes this course will have:          |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 1                                                                                         | Know the characteristics of various components.                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 2                                                                                         | Understand the utilization of components                                                                                                                                                                                                                                       |    |    |   |     |     |     |        |    |
| Course Outcomes: After completion of the course, the student will be able to              |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| S. No                                                                                     | Outcome                                                                                                                                                                                                                                                                        |    |    |   |     |     |     |        | KL |
| 1                                                                                         | Understand working of logic families and logic gates.                                                                                                                                                                                                                          |    |    |   |     |     |     |        | K2 |
| 2                                                                                         | Design and implement Combinational and Sequential logic circuits.                                                                                                                                                                                                              |    |    |   |     |     |     |        | K4 |
| 3                                                                                         | Solve elementary problems by assembly language programming                                                                                                                                                                                                                     |    |    |   |     |     |     |        | K3 |
| 4                                                                                         | Implement assembly language program for given task for 8086 microprocessor.                                                                                                                                                                                                    |    |    |   |     |     |     |        | K3 |
| SYLLABUS                                                                                  |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 1                                                                                         | Realization of Boolean Expressions using Gates                                                                                                                                                                                                                                 |    |    |   |     |     |     |        |    |
| 2                                                                                         | Design and realization logic gates using universal gates                                                                                                                                                                                                                       |    |    |   |     |     |     |        |    |
| 3                                                                                         | Designs a JK Flip-Flop, Edge triggered J-K NAND Flip Flop and show its functionality<br>Handle race condition and clock gating in your circuit.                                                                                                                                |    |    |   |     |     |     |        |    |
| 4                                                                                         | Design a 4 – bit Adder / Subtractor                                                                                                                                                                                                                                            |    |    |   |     |     |     |        |    |
| 5                                                                                         | Combinational logic circuits: Implementation of Boolean functions using logic gates                                                                                                                                                                                            |    |    |   |     |     |     |        |    |
| 6                                                                                         | Arithmetic operations using logic gates; Implementation of Multiplexers, Demultiplexers, Encoders, Decoders; Implementation of Boolean functions using Multiplexers/Decoders                                                                                                   |    |    |   |     |     |     |        |    |
| 7                                                                                         | Study of sequential logic circuits: Implementation of flip flops, Verify the excitation tables of various FLIP-FLOPS.                                                                                                                                                          |    |    |   |     |     |     |        |    |
| 8                                                                                         | Design and realization a Synchronous and Asynchronous counter using flip-flops                                                                                                                                                                                                 |    |    |   |     |     |     |        |    |
| 9                                                                                         | Design and realization of an 8-bit parallel load and serial out shift register using flip flops                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 10                                                                                        | Implementation of counters, Design and realization a Synchronous and Asynchronous counter using flip-flops                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 11                                                                                        | Design and realization of 4x1 mux, 8x1mux using 2x1 mux                                                                                                                                                                                                                        |    |    |   |     |     |     |        |    |
| Write assembly language programs in 8085/8086 for the following: (MASAM can also be used) |                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 1                                                                                         | To add two 8 bit number (A+B=RESULT with a carry and without a carry).                                                                                                                                                                                                         |    |    |   |     |     |     |        |    |
| 2                                                                                         | To subtract one 8 bit number from another (A-B=RESULT with a borrow and without a borrow).                                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 3                                                                                         | To find out AND, OR, NOT, XOR, NAND, NOR, XNOR of two 8 bit number                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 4                                                                                         | To find out addition of two 16 bit numbers.                                                                                                                                                                                                                                    |    |    |   |     |     |     |        |    |
| 5                                                                                         | To find out subtraction of two 16 bit numbers.                                                                                                                                                                                                                                 |    |    |   |     |     |     |        |    |
| 6                                                                                         | To evaluate the expression $a = b + c - d * e$ Considering 8-bit, 16 bit and 32-bit binary numbers as b, c, d, e. Take the input in consecutive memory locations and results also Display the results by using “int xx” of 8086. Validate program for the boundary conditions. |    |    |   |     |     |     |        |    |
| 7                                                                                         | To take N numbers as input. Perform the following operations on them                                                                                                                                                                                                           |    |    |   |     |     |     |        |    |

|                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                                                                                | a) Arrange in ascending and descending order.<br>b) Find max and minimum<br>c) Find average Considering 8-bit, 16-bit binary numbers and 2-digit, 4 digit and 8-digit BCD numbers. Display the results by using “int xx” of 8086. Validate program for the boundary conditions. |
| 8                                                                                                                                                                                                                                              | To implement the above operations as procedures and call from the main procedure                                                                                                                                                                                                |
| 9                                                                                                                                                                                                                                              | To find the factorial of a given number as a Procedure and call from the main program which display the result.                                                                                                                                                                 |
| Note: Experiments can be done using Logic board, EasyCPU, RTSlim, Little Man Computer (LMC), Assemblers for 8085 programming, 8086 based trainer kits, MIPS simulator PCSpim, Xilinx schematic editor and simulation tools or any other choice |                                                                                                                                                                                                                                                                                 |
| <b>e- Resources:</b>                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                 |
| 1.                                                                                                                                                                                                                                             | <a href="https://nptel.ac.in/courses/106/105/106105163/">https://nptel.ac.in/courses/106/105/106105163/</a>                                                                                                                                                                     |
| 2.                                                                                                                                                                                                                                             | <a href="http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf">http://www.cuc.ucc.ie/CS1101/David%20Tarnoff.pdf</a>                                                                                                                                                                 |

| Code                                                                                     | Category                                                                                                                                                                                                                                                                                                  | L  | T  | P | C   | I.M | E.M | Exam            |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|-----------------|
| B20CB2107                                                                                | PC                                                                                                                                                                                                                                                                                                        | -- | -- | 3 | 1.5 | 20  | 30  | 3 Hrs.          |
| OBJECT ORIENTED PROGRAMMING THROUGH JAVA LAB                                             |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| (For CSBS)                                                                               |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| Course Objectives: The aim of this lab is to                                             |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| 1.                                                                                       | Practice programming in the Java                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |                 |
| 2.                                                                                       | Gain knowledge of object-oriented paradigm in the Java programming language                                                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| 3.                                                                                       | Learn use of Java in a variety of technologies and on different platforms                                                                                                                                                                                                                                 |    |    |   |     |     |     |                 |
|                                                                                          |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| Course Outcomes: By the end of the course student will be able to write java program for |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| S.No                                                                                     | Outcome                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     | Knowledge Level |
| 1.                                                                                       | Apply primitive data types, Operations, Expressions, Control-flow, Strings in java programming                                                                                                                                                                                                            |    |    |   |     |     |     | K3              |
| 2.                                                                                       | Examine Class, Objects, Methods, Inheritance, Exception, Runtime Polymorphism, User defined Exception handling mechanism                                                                                                                                                                                  |    |    |   |     |     |     | K4              |
| 3.                                                                                       | Analyzing simple inheritance, multi-level inheritance, Exception handling mechanism                                                                                                                                                                                                                       |    |    |   |     |     |     | K4              |
| 4.                                                                                       | Analyze and Construct Threads, Event Handling, implement packages                                                                                                                                                                                                                                         |    |    |   |     |     |     | K4              |
|                                                                                          |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| LIST OF EXPERIMENTS                                                                      |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| Exercise - 1 (Basics)                                                                    |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| 1.                                                                                       | Write a JAVA program to display default value of all primitive data type of JAVA                                                                                                                                                                                                                          |    |    |   |     |     |     |                 |
| 2.                                                                                       | Write a java program that display the roots of a quadratic equation $ax^2+bx+c=0$ . Calculate the discriminate D and basing on value of D, describe the nature of root.                                                                                                                                   |    |    |   |     |     |     |                 |
| 3.                                                                                       | Five Bikers Compete in a race such that they drive at a constant speed which may or may not be the same as the other. To qualify the race, the speed of a racer must be more than the average speed of all 5 racers. Take as input the speed of each racer and print back the speed of qualifying racers. |    |    |   |     |     |     |                 |
| Exercise - 2 (Operations, Expressions, Control-flow, Strings)                            |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| 1.                                                                                       | Write a JAVA program to search for an element in a given list of elements using binary search mechanism.                                                                                                                                                                                                  |    |    |   |     |     |     |                 |
| 2.                                                                                       | Write a JAVA program to sort for an element in a given list of elements using bubble sort                                                                                                                                                                                                                 |    |    |   |     |     |     |                 |
| 3.                                                                                       | Write a JAVA program to sort for an element in a given list of elements using merge sort.                                                                                                                                                                                                                 |    |    |   |     |     |     |                 |
| 4.                                                                                       | Write a JAVA program using StringBuffer to delete, remove character.                                                                                                                                                                                                                                      |    |    |   |     |     |     |                 |
| Exercise - 3 (Class, Objects)                                                            |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| 1.                                                                                       | Write a JAVA program to implement class mechanism. – Create a class, methods and invoke them inside main method.                                                                                                                                                                                          |    |    |   |     |     |     |                 |
| 2.                                                                                       | Write a JAVA program to implement constructor.                                                                                                                                                                                                                                                            |    |    |   |     |     |     |                 |
| Exercise - 4 (Methods)                                                                   |                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |                 |
| 1.                                                                                       | Write a JAVA program to implement constructor overloading.                                                                                                                                                                                                                                                |    |    |   |     |     |     |                 |
| 2.                                                                                       | Write a JAVA program implement method overloading.                                                                                                                                                                                                                                                        |    |    |   |     |     |     |                 |

|                                               |                                                                                                                                                                                                                                                                      |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                               |                                                                                                                                                                                                                                                                      |
| <b>Exercise - 5 (Inheritance)</b>             |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program to implement Single Inheritance                                                                                                                                                                                                                 |
| 2.                                            | Write a JAVA program to implement multi level Inheritance                                                                                                                                                                                                            |
| 3.                                            | Write a java program for abstract class to find areas of different shapes                                                                                                                                                                                            |
| <b>Exercise - 6 (Inheritance - Continued)</b> |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program give example for “super” keyword.                                                                                                                                                                                                               |
| 2.                                            | Write a JAVA program to implement Interface. What kind of Inheritance can be achieved?                                                                                                                                                                               |
| <b>Exercise - 7 (Exception)</b>               |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program that describes exception handling mechanism                                                                                                                                                                                                     |
| 2.                                            | Write a JAVA program Illustrating Multiple catch clauses                                                                                                                                                                                                             |
| <b>Exercise – 8 (Runtime Polymorphism)</b>    |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program that implements Runtime polymorphism                                                                                                                                                                                                            |
| 2.                                            | Write a Case study on run time polymorphism, inheritance that implements in above problem                                                                                                                                                                            |
| <b>Exercise – 9 (User defined Exception)</b>  |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program for creation of Illustrating throw                                                                                                                                                                                                              |
| 2.                                            | Write a JAVA program for creation of Illustrating finally                                                                                                                                                                                                            |
| 3.                                            | Write a JAVA program for creation of Java Built-in Exceptions                                                                                                                                                                                                        |
| 4.                                            | Write a JAVA program for creation of User Defined Exception                                                                                                                                                                                                          |
| <b>Exercise – 10 (Threads)</b>                |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program that creates threads by extending Thread class .First thread display “Good Morning “every 1 sec, the second thread displays “Hello “every 2 seconds and the third display “Welcome” every 3 seconds ,(Repeat the same by implementing Runnable) |
| 2.                                            | Write a program illustrating <b>isAlive</b> and <b>join ()</b>                                                                                                                                                                                                       |
| <b>Exercise – 11 (Packages)</b>               |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program illustrate class path                                                                                                                                                                                                                           |
| 2.                                            | Write a case study on including in class path in your OS environment of your package.                                                                                                                                                                                |
| 3.                                            | Write a JAVA program that import and use the defined your package in the previous Problem                                                                                                                                                                            |
| <b>Exercise – 12 (Event Handling)</b>         |                                                                                                                                                                                                                                                                      |
| 1.                                            | Write a JAVA program that display the x and y position of the cursor movement using Mouse.                                                                                                                                                                           |
| 2.                                            | Write a Java program to create radio buttons(male & female) perform event handling to display relevant text when radio button selected and button press is performed.                                                                                                |
| 3.                                            | Write a java program to Demonstrate KeyAdapter classes.                                                                                                                                                                                                              |
| <b>Reference Books:</b>                       |                                                                                                                                                                                                                                                                      |
| 1.                                            | JAVA one step ahead, Anitha Seth, B.L.Juneja, Oxford.                                                                                                                                                                                                                |
| 2.                                            | The complete Reference Java, 8th edition, Herbert Schildt, TMH.                                                                                                                                                                                                      |

| Code                                                                               | Category                                                                                                                                                                                                                                                                                                 | L | T  | P | C | I.M | E.M | Exam   |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|-----|-----|--------|
| B20IT2108                                                                          | SOC                                                                                                                                                                                                                                                                                                      | - | -- | 4 | 2 | --  | 50  | 3 Hrs. |
| NETWORK ADMINISTRATION                                                             |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| (Skill Oriented Course-I)                                                          |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| (Common to AIDS ,CSBS & IT)                                                        |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| Course Objectives: On completing this course student will be able to               |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| 1                                                                                  | Install different Operating Systems, antivirus and components                                                                                                                                                                                                                                            |   |    |   |   |     |     |        |
| 2                                                                                  | Install and configure windows server.                                                                                                                                                                                                                                                                    |   |    |   |   |     |     |        |
| 3                                                                                  | Install and configure different networking protocols and network tools.                                                                                                                                                                                                                                  |   |    |   |   |     |     |        |
| Course Outcomes: By the end of the course, the student should have the ability to: |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| S.No                                                                               | Outcome                                                                                                                                                                                                                                                                                                  |   |    |   |   |     |     | KL     |
| 1                                                                                  | Demonstrate installation and configuration of Operating systems.                                                                                                                                                                                                                                         |   |    |   |   |     |     | K2     |
| 2                                                                                  | Demonstrate installation and configuration of DNS, DHCP, set up adhoc networks.                                                                                                                                                                                                                          |   |    |   |   |     |     | K2     |
| 3                                                                                  | Demonstrate different network related tools and applications.                                                                                                                                                                                                                                            |   |    |   |   |     |     | K2     |
| SYLLABUS                                                                           |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| 1                                                                                  | Install MS Windows from the CD/DVD. We can try to install version from 10 to XP (8.1 recommended). Installation should include users, administrators, device configuration, patches and updates, antivir system, necessary environment configuration (i.e. NumLock, hiding of extension settings, etc.). |   |    |   |   |     |     |        |
| 2                                                                                  | HW installation. Changing harddrives, preparing cables, setting of a separated network (course on real computers, harddrives dedicated for the course allows hardware sharing).                                                                                                                          |   |    |   |   |     |     |        |
| 3                                                                                  | Install selected Linux distribution (Debian or Slackware recommended). Settings, as for MS Windows.                                                                                                                                                                                                      |   |    |   |   |     |     |        |
| 4                                                                                  | Text mode console commands. Cron, scripts.                                                                                                                                                                                                                                                               |   |    |   |   |     |     |        |
| 5                                                                                  | Novell Netware administration, if anyone interested. Novell is not used for many years, but is useful to describe, how the server can work                                                                                                                                                               |   |    |   |   |     |     |        |
| 6                                                                                  | Work with server - SQL, Apache, FTP/SFTP, mail. Mail robots.                                                                                                                                                                                                                                             |   |    |   |   |     |     |        |
| 7                                                                                  | Server as a firewall, ip sharing (NAT, DNAT, masquerade). Server behind the firewall.                                                                                                                                                                                                                    |   |    |   |   |     |     |        |
| 8                                                                                  | Installation & configuration of Windows professional.                                                                                                                                                                                                                                                    |   |    |   |   |     |     |        |
| 9                                                                                  | Installation & configuration of Windows server.                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| 10                                                                                 | Installing and Configuring Terminal Services                                                                                                                                                                                                                                                             |   |    |   |   |     |     |        |
| 11                                                                                 | Installing DNS. Implementing DNS in windows networks.                                                                                                                                                                                                                                                    |   |    |   |   |     |     |        |
| 12                                                                                 | Installing and configuring DHCP.                                                                                                                                                                                                                                                                         |   |    |   |   |     |     |        |
| 13                                                                                 | Configuring & Implementing routing services                                                                                                                                                                                                                                                              |   |    |   |   |     |     |        |
| 14                                                                                 | Configuration and setup of adhoc network and infrastructure network.                                                                                                                                                                                                                                     |   |    |   |   |     |     |        |
| 15                                                                                 | NET tools, Deployment of NETTOOLS.                                                                                                                                                                                                                                                                       |   |    |   |   |     |     |        |
| 16                                                                                 | Tracing of email origin using email trace pro utility.                                                                                                                                                                                                                                                   |   |    |   |   |     |     |        |
| 17                                                                                 | Use of key logges and anti key logger to secure your system                                                                                                                                                                                                                                              |   |    |   |   |     |     |        |
| Reference Books:                                                                   |                                                                                                                                                                                                                                                                                                          |   |    |   |   |     |     |        |
| 1                                                                                  | Mastering Windows 2000 Server by Mark Minasi, Sybex, ISBN: 0-7821-2774-6                                                                                                                                                                                                                                 |   |    |   |   |     |     |        |
| 2                                                                                  | Computer Networks- Andrew S Tanenbaum, 4 <sup>th</sup> edition, Pearson Education                                                                                                                                                                                                                        |   |    |   |   |     |     |        |

| Code                                                                               | Category                                                                                                   | L | T  | P | C | I.M | E.M | Exam            |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|---|----|---|---|-----|-----|-----------------|
| B20IT2109                                                                          | SOC                                                                                                        | - | -- | 4 | 2 | --  | 50  | 3 Hrs.          |
|                                                                                    |                                                                                                            |   |    |   |   |     |     |                 |
| SPREAD SHEET DATA ANALYSIS                                                         |                                                                                                            |   |    |   |   |     |     |                 |
| (Skill Oriented Course-I)                                                          |                                                                                                            |   |    |   |   |     |     |                 |
| (Common to AIDS ,CSBS & IT)                                                        |                                                                                                            |   |    |   |   |     |     |                 |
| Course Objectives: On completing this course student will be able to               |                                                                                                            |   |    |   |   |     |     |                 |
| 1                                                                                  | To develop basic knowledge in Excel                                                                        |   |    |   |   |     |     |                 |
| 2                                                                                  | To expose the various functions in Excel                                                                   |   |    |   |   |     |     |                 |
| 3                                                                                  | To extend the skill to use data visualization                                                              |   |    |   |   |     |     |                 |
| 4                                                                                  | To analyze the real time datasets                                                                          |   |    |   |   |     |     |                 |
| 5                                                                                  | To develop Pivot tables and VLOOKUP functions                                                              |   |    |   |   |     |     |                 |
|                                                                                    |                                                                                                            |   |    |   |   |     |     |                 |
| Course Outcomes: By the end of the course, the student should have the ability to: |                                                                                                            |   |    |   |   |     |     |                 |
| S.No                                                                               | Outcome                                                                                                    |   |    |   |   |     |     | Knowledge Level |
| 1                                                                                  | Describe common Excel functionality and features used for data science                                     |   |    |   |   |     |     | K2              |
| 2                                                                                  | Analyze and construct the data Visualization                                                               |   |    |   |   |     |     | K3              |
| 3                                                                                  | Configure the programming environment                                                                      |   |    |   |   |     |     | K2              |
| 4                                                                                  | Analyze real time data set                                                                                 |   |    |   |   |     |     | K3              |
| 5                                                                                  | Implement Pivot tables and VLOOKUP functions                                                               |   |    |   |   |     |     | K3              |
|                                                                                    |                                                                                                            |   |    |   |   |     |     |                 |
| SYLLABUS                                                                           |                                                                                                            |   |    |   |   |     |     |                 |
| List of experiments                                                                |                                                                                                            |   |    |   |   |     |     |                 |
| 1                                                                                  | Study of basic functions in Excel                                                                          |   |    |   |   |     |     |                 |
| 2                                                                                  | Working with Range Names and Tables                                                                        |   |    |   |   |     |     |                 |
| 3                                                                                  | Cleaning Data with Text Functions.                                                                         |   |    |   |   |     |     |                 |
| 4                                                                                  | Working with VLOOKUP functions and Pivot tables.                                                           |   |    |   |   |     |     |                 |
| 5                                                                                  | Demonstration of data Visualization                                                                        |   |    |   |   |     |     |                 |
| 6                                                                                  | Importing data from external source into excel                                                             |   |    |   |   |     |     |                 |
| 7                                                                                  | Creating a data model                                                                                      |   |    |   |   |     |     |                 |
| 8                                                                                  | Exploring data with Pivot tables                                                                           |   |    |   |   |     |     |                 |
| 9                                                                                  | Create a dash board for a given requirement.                                                               |   |    |   |   |     |     |                 |
| 10                                                                                 | Implement a data analytics for the real time data set.                                                     |   |    |   |   |     |     |                 |
| Reference Books:                                                                   |                                                                                                            |   |    |   |   |     |     |                 |
| 1.                                                                                 | Julio Cesar Rodriques Martino, “ Hands – on Machine learning with Microsoft Excel”, Packt Publication 2019 |   |    |   |   |     |     |                 |
| 2.                                                                                 | Paul McFedries, “Excel data analysis for dunnies”, john wiley and sone 2019.                               |   |    |   |   |     |     |                 |

| Code                                                                       | Category                                                                                                                                                                                                                                                                                                                                               | L | T  | P  | C | I.M | E.M | Exam |
|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|-----|------|
| B20MC2102                                                                  | MC                                                                                                                                                                                                                                                                                                                                                     | 2 | -- | -- | 0 | --  | --  | --   |
| PROFESSIONAL ETHICS & HUMAN VALUES                                         |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| (Common to CE,EEE,ME,AIDS & CSBS)                                          |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| Course Objectives: The objectives of the course are to impart:             |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| 1.                                                                         | To create an awareness on Engineering Ethics and Human Values.                                                                                                                                                                                                                                                                                         |   |    |    |   |     |     |      |
| 2.                                                                         | To instill Moral and Social Values and Loyalty.                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| 3.                                                                         | To appreciate the rights of others.                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| Course outcomes : After completion of the course, students will be able to |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     | K L  |
| 1                                                                          | Identify and analyze an ethical issue in the subject matter under investigation or in a relevant field.                                                                                                                                                                                                                                                |   |    |    |   |     |     | K3   |
| 2                                                                          | Identify the multiple ethical interests at stake in a real-world situation or practice.                                                                                                                                                                                                                                                                |   |    |    |   |     |     | K2   |
| 3                                                                          | Articulate what makes a particular course of action ethically defensible.                                                                                                                                                                                                                                                                              |   |    |    |   |     |     | K3   |
| 4                                                                          | Assess their own ethical values and the social context of problems.                                                                                                                                                                                                                                                                                    |   |    |    |   |     |     | K3   |
| 5                                                                          | Identify ethical concerns in research and intellectual contexts, including academic integrity, use and citation of sources, the objective presentation of data, and the treatment of human subjects.                                                                                                                                                   |   |    |    |   |     |     | K2   |
| 6                                                                          | Demonstrate knowledge of ethical values in non-classroom activities, such as service learning, internships, and field work.                                                                                                                                                                                                                            |   |    |    |   |     |     | K3   |
| 7                                                                          | Integrate, synthesize, and apply knowledge of ethical dilemmas and resolutions in academic settings, including focused and interdisciplinary research.                                                                                                                                                                                                 |   |    |    |   |     |     | K4   |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| SYLLABUS                                                                   |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| UNIT-I<br>(8 Hrs)                                                          | Human Values: Morals, Values and Ethics-Integrity-Work Ethic-Service learning Civic Virtue Respect for others Living Peacefully Caring Sharing Honesty -Courage-Cooperation Commitment Empathy Self Confidence Character Spirituality..                                                                                                                |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| UNIT-II<br>(8 Hrs)                                                         | Engineering Ethics: Senses of 'Engineering Ethics-Variety of moral issued- Types of inquiry Moral dilemmas Moral autonomy- Kohlberg's theory- Gilligan's theory- Consensus and controversy Models of professional roles-Theories about right action-Self-interest -Customs and religion Uses of Ethical theories Valuing time Cooperation Commitment., |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| UNIT-III<br>(8 Hrs)                                                        | Engineering as Social Experimentation: Engineering As Social Experimentation-Framing the problem- Determining the facts codes of Ethics-Clarifying Concepts-Application issues Common Ground -General Principles- Utilitarian thinking respect for persons                                                                                             |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |
| UNIT-IV<br>(8 Hrs)                                                         | Engineers Responsibility for Safety and Risk: Safety and risk Assessment of safety and risk. Risk benefit analysis and reducing risk-Safety and the Engineer-Designing for the safety- Intellectual Property rights(IPR).,                                                                                                                             |   |    |    |   |     |     |      |
|                                                                            |                                                                                                                                                                                                                                                                                                                                                        |   |    |    |   |     |     |      |

|                           |                                                                                                                                                                                                                                                                                                                             |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-V<br/>(8 Hrs)</b> | <b>Global Issues:</b> Globalization- Cross-culture issues-Environmental Ethics- Computer Ethics Computers as the instrument of Unethical behavior Computers as the object of Unethical acts Autonomous Computers-Computer codes of Ethics- Weapons Development -Ethics and Research Analyzing Ethical Problems in research. |
| <b>Text Books:</b>        |                                                                                                                                                                                                                                                                                                                             |
| 1.                        | Engineering Ethics includes Human Values" by M.Govindarajan, S.Natarajan- and, V.S.Senthil Kumar-PHI Learning Pvt Ltd-2009.                                                                                                                                                                                                 |
| 2.                        | "Engineering Ethics" by Harris, Pritchard and Rabins, CENGAGE Learning, India Edition, 2009.                                                                                                                                                                                                                                |
| 3.                        | "Ethics in Engineering" by Mike W. Martin and Roland Schinzinger - Tata McGraw-Hill-2003.                                                                                                                                                                                                                                   |
| 4.                        | "Professional Ethics and Morals" by Prof.A.R.Aryasri, DhanikotaSuyodhana-Maruthi Publications.                                                                                                                                                                                                                              |
| 5.                        | "Professional Ethics and Human Values" by A.Alavudeen, R.Kalil Rahman and M.Jayakumaran-LaxmiPublications.                                                                                                                                                                                                                  |
| 6.                        | "Professional Ethics and Human Values" by Prof.D.R.Kiran-                                                                                                                                                                                                                                                                   |
| 7.                        | "Indian Culture, Values and Professional Ethics" by PSR Murthy- BS Publication.                                                                                                                                                                                                                                             |
| 8.                        | Professional Ethics by R.Subramaniam - Oxford publications, New Delhi                                                                                                                                                                                                                                                       |
|                           |                                                                                                                                                                                                                                                                                                                             |

| Regulation: R20                                                                          |                                            |          | II / IV - B.Tech. II - Semester |    |   |    |            |            |             |
|------------------------------------------------------------------------------------------|--------------------------------------------|----------|---------------------------------|----|---|----|------------|------------|-------------|
| COMPUTER SCIENCE & BUSINESS SYSTEM                                                       |                                            |          |                                 |    |   |    |            |            |             |
| SCHEME OF INSTRUCTION & EXAMINATION<br>(With effect from 2020-21 admitted Batch onwards) |                                            |          |                                 |    |   |    |            |            |             |
| Course Code                                                                              | Course Name                                | Category | Cr                              | L  | T | P  | Int. Marks | Ext. Marks | Total Marks |
| B20 CB 2201                                                                              | Operating Systems                          | PC       | 3                               | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2202                                                                              | Python Programming                         | PC       | 3                               | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2203                                                                              | Design and Analysis of Algorithms          | PC       | 3                               | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 HS 2202                                                                              | Fundamentals of Management                 | PC       | 3                               | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 HS 2203                                                                              | Business Communication & Value Science III | HS       | 3                               | 3  | 0 | 0  | 30         | 70         | 100         |
| B20 CB 2204                                                                              | OS & Unix Programming Lab                  | PC       | 1.5                             | 0  | 0 | 3  | 15         | 35         | 50          |
| B20 CB 2205                                                                              | Python Programming Lab                     | PC       | 1.5                             | 0  | 0 | 3  | 15         | 35         | 50          |
| B20 CB 2206                                                                              | Design and Analysis of Algorithms Lab      | PC       | 1.5                             | 0  | 0 | 3  | 15         | 35         | 50          |
| #SOC-II                                                                                  | Skill Oriented Course-II                   | SOC      | 2                               | 0  | 0 | 4  | --         | 50         | 50          |
| B20 MC 2201                                                                              | English Proficiency                        | MC       | 0                               | 2  | 0 | 0  | --         | --         | --          |
| TOTAL                                                                                    |                                            |          | 21.5                            | 17 | 0 | 13 | 210        | 490        | 700         |

| #SOC-II | Course Code | Name of the Course   |
|---------|-------------|----------------------|
|         | B20 IT 2207 | Animations           |
|         | B20 IT 2208 | Web Design Using PHP |

| Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Category                                                                                                                                       | L | T  | P  | C | I.M | E.M | Exam   |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|---|-----|-----|--------|-----|
| B20CB2201                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | PC                                                                                                                                             | 3 | -- | -- | 3 | 30  | 70  | 3 Hrs. |     |
| OPERATING SYSTEMS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                |   |    |    |   |     |     |        |     |
| (For CSBS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                |   |    |    |   |     |     |        |     |
| Course Objectives: Students are expected to learn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                |   |    |    |   |     |     |        |     |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Introduce to the internal operation of modern operating systems                                                                                |   |    |    |   |     |     |        |     |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Define, explain, processes and threads, mutual exclusion, CPU scheduling, deadlock, memory management, and file systems                        |   |    |    |   |     |     |        |     |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Understand File Systems in Operating System like UNIX/Linux and Windows                                                                        |   |    |    |   |     |     |        |     |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Understand Input Output Management and use of Device Driver and Secondary Storage (Disk)Mechanism                                              |   |    |    |   |     |     |        |     |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Analyze Security and Protection Mechanism in Operating System                                                                                  |   |    |    |   |     |     |        |     |
| Course Outcomes:After completion of the course, the student will be able to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                |   |    |    |   |     |     |        |     |
| S.No                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Outcome                                                                                                                                        |   |    |    |   |     |     |        | K L |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Describe various generations of Operating System and functions of Operating System                                                             |   |    |    |   |     |     |        | K2  |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Describe the concept of program, process and thread and analyze various CPU Scheduling Algorithms and compare their performance                |   |    |    |   |     |     |        | K3  |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Solve Inter Process Communication problems using Mathematical Equations by various methods                                                     |   |    |    |   |     |     |        | K3  |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Compare various Memory Management Schemes especially paging and Segmentation in Operating System and apply various Page Replacement Techniques |   |    |    |   |     |     |        | K3  |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Outline the basic functionality of Operating System like UNIX/Linux and Windows.                                                               |   |    |    |   |     |     |        | K2  |
| SYLLABUS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                |   |    |    |   |     |     |        |     |
| UNIT-I (12Hrs)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                |   |    |    |   |     |     |        |     |
| Operating Systems Overview: Operating system functions, Operating system structure, Operating systems operations, Computing environments, Open-Source Operating Systems.<br>System Structures: Operating System Services, User and Operating-System Interface, systems calls, Types of System Calls, system programs, operating system structure, operating system debugging, System Boot.                                                                                                                                                                                                                                            |                                                                                                                                                |   |    |    |   |     |     |        |     |
| UNIT-II (10Hrs)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                |   |    |    |   |     |     |        |     |
| Process Concept: Process scheduling, Operations on processes, Inter-process communication, Communication in client server systems.<br>Multithreaded Programming: Multithreading models, Thread libraries, Threading issues.<br>Process Scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple processor scheduling, Thread scheduling.<br>Inter-process Communication: Race conditions, Critical Regions, Mutual exclusion with Busy waiting, leep and wakeup, Semaphores, Mutexes, Monitors, Message passing, Barriers, Classical IPC Problems - Dining philosophers problem, Readers and writers problem. |                                                                                                                                                |   |    |    |   |     |     |        |     |

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| <b>UNIT-III<br/>(14Hrs)</b>                                                         | <b>Memory-Management Strategies:</b> Introduction, Swapping, Contiguous memory allocation, Paging, Segmentation.<br><b>Virtual Memory Management:</b> Introduction, Demand paging, Copy on-write, Page replacement, Frame allocation, Thrashing, Memory-mapped files, Kernel memory allocation.                                                                                                                     |
| <b>UNIT-IV<br/>(10Hrs)</b>                                                          | <b>Deadlocks:</b> Resources, Conditions for resource deadlocks, Ostrich algorithm, Deadlock detection and recovery, Deadlock avoidance, Deadlock prevention.<br><b>File Systems:</b> Files, Directories, File system implementation, management and optimization.<br><b>Secondary-Storage Structure:</b> Overview of disk structure, and attachment, Disk scheduling, RAID structure, Stable storage implementation |
| <b>UNIT-V<br/>(12Hrs)</b>                                                           | <b>System Protection:</b> Goals of protection, Principles and domain of protection, Access matrix, Access control, Revocation of access rights.<br><b>System Security:</b> Introduction, Program threats, System and network threats, Cryptography for security, User authentication, Implementing security defenses, Firewalling to protect systems and networks, Computer security classification.                |
| <b>Note: discuss the overview of Linux and Microsoft Windows operating systems.</b> |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Text Books:</b>                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.                                                                                  | Silberschatz A, Galvin P B, and Gagne G, Operating System Concepts, 9th edition, Wiley, 2013.                                                                                                                                                                                                                                                                                                                       |
| 2.                                                                                  | Tanenbaum A S, Modern Operating Systems, 3rd edition, Pearson Education, 2008. (for Inter Process Communication and File Systems.)                                                                                                                                                                                                                                                                                  |
| <b>Reference Books:</b>                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.                                                                                  | Dhamdhare D M, Operating Systems A Concept Based Approach, 3rd edition, Tata McGraw-Hill, 2012.                                                                                                                                                                                                                                                                                                                     |
| 2                                                                                   | Stallings W, Operating Systems -Internals and Design Principles, 6th edition, Pearson Education,2009                                                                                                                                                                                                                                                                                                                |
| 3                                                                                   | Nutt G, Operating Systems, 3rd edition, Pearson Education,2004                                                                                                                                                                                                                                                                                                                                                      |
| <b>e-Resources:</b>                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1                                                                                   | <a href="https://nptel.ac.in/courses/106/105/106105214/">https://nptel.ac.in/courses/106/105/106105214/</a>                                                                                                                                                                                                                                                                                                         |

| Course Code                                                                  | Category                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | L | T  | P  | C | I.M | E.M. | Exam   |
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| B20CB2202                                                                    | PC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 3 | -- | -- | 3 | 30  | 70   | 3 Hrs. |
| PYTHON PROGRAMMING                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |      |        |
| (For CSBS)                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |      |        |
| Course Objectives: Students are expected to learn                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |      |        |
| 1.                                                                           | To learn about Python programming language syntax, semantics, and the runtime environment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |    |    |   |     |      |        |
| 2.                                                                           | To be familiarized with universal computer programming concepts like data types, containers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |
| 3.                                                                           | To be familiarized with general computer programming concepts like conditional execution, loops & functions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |    |   |     |      |        |
| 4.                                                                           | To be familiarized with general coding techniques and object-oriented programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |      |        |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |      |        |
| S.No                                                                         | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |    |    |   |     |      | K L    |
| 1                                                                            | Develop essential programming skills in computer programming concepts like data types, containers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |    |    |   |     |      | K4     |
| 2                                                                            | Apply the basics of programming in the Python language                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |    |    |   |     |      | K3     |
| 3                                                                            | Solve coding tasks related conditional execution, loops                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |    |    |   |     |      | K3     |
| 4                                                                            | Solve coding tasks related to the fundamental notions and techniques used in object-oriented programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |    |    |   |     |      | K3     |
| 5                                                                            | Implement user define exception and GUI application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |    |    |   |     |      | K3     |
| SYLLABUS                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |    |    |   |     |      |        |
| UNIT-I<br>(10Hrs)                                                            | Introduction: Introduction to Python, Program Development Cycle, Input, Processing, and Output, Displaying Output with the Print Function, Comments, Variables, Reading Input from the Keyboard, Performing Calculations, Operators. Type conversions, Expressions, More about Data Output.<br>Data Types, and Expression: Strings Assignment, and Comment, Numeric Data Types and Character Sets, Using functions and Modules.<br>Decision Structures and Boolean Logic: if, if-else, if-elif-else Statements, Nested Decision Structures, Comparing Strings, Logical Operators, Boolean Variables. |   |    |    |   |     |      |        |
| UNIT-II<br>(08Hrs)                                                           | Repetition Structures: Introduction, while loop, for loop, Calculating a Running Total, Input Validation Loops, Nested Loops.<br>Control Statement: Definite iteration for Loop Formatting Text for output.<br>Strings and Text Files: Accessing Character and Substring in Strings, Data Encryption, Strings and Number Systems, String Methods Text Files.                                                                                                                                                                                                                                         |   |    |    |   |     |      |        |
| UNIT-III<br>(10Hrs)                                                          | List and Dictionaries: Lists, Defining Simple Functions, Dictionaries<br>Design with Function: Functions as Abstraction Mechanisms, Problem Solving with Top Down Design, Design with Recursive Functions, Case Study Gathering Information from a File System, Managing a Program’s Namespace, Higher Order Function.<br>Modules: Modules, Standard Modules, Packages.                                                                                                                                                                                                                              |   |    |    |   |     |      |        |

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| <b>UNIT-IV<br/>(10Hrs)</b>  | <p><b>File Operations:</b> Reading config files in python, Writing log files in python, Understanding read functions, read(), readline() and readlines(), Understanding write functions, write() and writelines(), Manipulating file pointer using seek, Programming using file operations</p> <p><b>Object Oriented Programming:</b> Concept of class, object and instances, Constructor, class attributes and destructors, Real time use of class in live projects, Inheritance , overlapping and overloading operators, Adding and retrieving dynamic attributes of classes, Programming using Oops support</p> <p><b>Design with Classes:</b> Objects and Classes, Data modeling Examples, Case Study An ATM, Structuring Classes with Inheritance and Polymorphism</p> |
| <b>UNIT-V<br/>(12 Hrs.)</b> | <p><b>Errors and Exceptions:</b> Syntax Errors, Exceptions, Handling Exceptions, Raising Exceptions, User-defined Exceptions, Defining Clean-up Actions, Redefined Clean-up Actions.</p> <p><b>Graphical User Interfaces:</b> The Behavior of Terminal Based Programs and GUI -Based, Programs, Coding Simple GUI-Based Programs, Other Useful GUI Resources.</p> <p><b>Programming:</b> Introduction to Programming Concepts with Scratch.</p>                                                                                                                                                                                                                                                                                                                             |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.                          | Fundamentals of Python First Programs, Kenneth. A. Lambert, Cengage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2.                          | Python Programming: A Modern Approach, Vamsi Kurama, Pearson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 1.                          | Introduction to Python Programming, Gowrishankar.S, Veena A, CRC Press.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2.                          | Programming and Problem Solving with Python, Ashok Namdev Kamthane, Amit Ashok Kamthane, TMH, 2019.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3.                          | Introduction to Programming Using Python, Y. Daniel Liang, Pearson.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 4.                          | <a href="https://www.tutorialspoint.com/python3/python_tutorial.pdf">https://www.tutorialspoint.com/python3/python_tutorial.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Subject Code                                                                       | Category                                                                                                                                                                                      | L                                                                                                                                                                                                                                                                                                                      | T  | P  | C | I.M | E.M | Exam            |
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| B20CB2203                                                                          | PC                                                                                                                                                                                            | 3                                                                                                                                                                                                                                                                                                                      | -- | -- | 3 | 25  | 75  | 3 Hrs.          |
| DESIGN AND ANALYSIS OF ALGORITHMS                                                  |                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| Course Objectives: On completing this course student will be able to               |                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| 1                                                                                  | Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| 2                                                                                  | Analyze the asymptotic performance of algorithms.                                                                                                                                             |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| 3                                                                                  | Demonstrate a familiarity with major algorithms and data structures                                                                                                                           |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| 4                                                                                  | Synthesize efficient algorithms in common engineering design situations.                                                                                                                      |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| Course Outcomes: By the end of the course, the student should have the ability to: |                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| S.No                                                                               | Outcome                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | Knowledge Level |
| 1                                                                                  | Apply the mathematical principle to analyze the efficiency of algorithms by measuring time complexity & space complexity.                                                                     |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | K3              |
| 2                                                                                  | Apply the Divide-and-Conquer strategy and Greedy Method for solving the complex problems & analyze the performance of solutions.                                                              |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | K3              |
| 3                                                                                  | Apply the optimistic strategies Dynamic Programming for computational problems in computer field.                                                                                             |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | K3              |
| 4                                                                                  | Apply the Backtracking and Branch-and-bound strategies for solving complex problems                                                                                                           |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | K3              |
| 5                                                                                  | Understand the basic concepts of NP-Hard and NP- Complete and Solve string matching using various algorithms.                                                                                 |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     | K2              |
| SYLLABUS                                                                           |                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |                 |
| UNIT-I<br>(10 Hrs)                                                                 |                                                                                                                                                                                               | Introduction: Algorithm Definition, Algorithm Specification, performance Analysis, Randomized Algorithms. Sets & Disjoint set union: introduction, union and find operations.<br>Basic Traversal & Search Techniques: Techniques for Graphs, connected components and Spanning Trees, Bi-connected components and DFS. |    |    |   |     |     |                 |
| UNIT-II<br>(10 Hrs)                                                                |                                                                                                                                                                                               | Divide and Conquer: General Method, Defective chessboard, Binary Search, finding the maximum and minimum, Merge sort, Quick sort.<br>The Greedy Method: The general Method, container loading, knapsack problem, Job sequencing with deadlines, minimum cost spanning Trees.                                           |    |    |   |     |     |                 |
| UNIT-III<br>(10 Hrs)                                                               |                                                                                                                                                                                               | Dynamic Programming: The general method, multistage graphs, All pairs-shortest paths, single-source shortest paths: general weights, optimal Binary search trees, 0/1 knapsack, reliability Design, The traveling salesperson problem                                                                                  |    |    |   |     |     |                 |

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| <b>UNIT-IV<br/>(8 Hrs)</b> | <b>Backtracking:</b> The General Method, The 8-Queens problem, sum of subsets, Graph coloring, Hamiltonian cycles, knapsack problem.<br><b>Branch and Bound:</b> FIFO Branch-and-Bound, LC Branch-and-Bound, 0/1 Knapsack problem, Traveling salesperson problem.  |
| <b>UNIT-V<br/>(12 Hrs)</b> | <b>NP-Hard and NP-Complete problems:</b> Basic concepts, Cook's Theorem.<br><b>String Matching:</b> Introduction, String Matching-Meaning and Application, Naïve String Matching Algorithm, Rabin-Karp Algorithm, Knuth-Morris-Pratt Automata, Tries, Suffix Tree. |
| <b>Text Books:</b>         |                                                                                                                                                                                                                                                                    |
| 1.                         | Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "Fundamentals of Computer Algorithms", 2nd Edition, Universities Press.                                                                                                                                     |
| 2.                         | Harsh Bhasin, "Algorithms Design & Analysis", Oxford University Press.                                                                                                                                                                                             |
| <b>Reference Books:</b>    |                                                                                                                                                                                                                                                                    |
| 1.                         | S. Sridhar, "Design and Analysis of Algorithms", Oxford University Press.                                                                                                                                                                                          |

| Code                                                                         | Category                                                                                                               | L                                                                                                                                                                                                                                                                                                                                                                                                                                                       | T  | P  | C | I.M | E.M | Exam   |
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| B20HS2202                                                                    | PC                                                                                                                     | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                       | -- | -- | 3 | 30  | 70  | 3 Hrs. |
| FUNDAMENTALS OF MANAGEMENT                                                   |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| (For CSBS)                                                                   |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| Course Objectives: Students are expected to                                  |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| 1                                                                            | To familiarize with the concept of management, functions and principles                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| 2                                                                            | To provide conceptual knowledge on functional management that is on Human resource management and Marketing management |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| 3                                                                            | To provide basic insight into financial aspects and Strategic Management                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| 4                                                                            | To learn theories of motivation and deals with individual behavior, their attitude and perception of individuals       |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| 5                                                                            | To understand about Managerial Ethics in Business organizations and know how to manage employee stress                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| Course Outcomes: After completion of the course, the student will be able to |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| S. No                                                                        | Outcome                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K L    |
| 1                                                                            | Explain management functions and principles                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| 2                                                                            | Describe the concepts of functional management that is HRM and Marketing functions                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| 3                                                                            | Discuss about capital and methods to raise finance                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| 4                                                                            | Discuss about vision, mission, goal, objective and a strategy based on which the corporate planning depends            |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| 5                                                                            | Discuss about Individual behaviour and motivational theories                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| 6                                                                            | Describe about Ethics in Business and ways in managing stress                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     | K2     |
| SYLLABUS                                                                     |                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |        |
| UNIT-I<br>(10 Hrs)                                                           |                                                                                                                        | Introduction to Management:<br>Management: Concept, Nature and importance of Management, Evolution of Management thought, Contribution of Management Thinkers - Taylor's Scientific Management&Fayol's principles of Management, Functions of Management - Planning, Organising, Staffing, Directing & Controlling                                                                                                                                      |    |    |   |     |     |        |
| UNIT-II<br>(10 Hrs)                                                          |                                                                                                                        | Functional Management:<br>Human Resource Management(HRM): Concept, Nature & Importance of HRM,Origin and Transmission of HRM, Basic functions of HR Manager: Manpower planning, Recruitment, Selection, Training and Development, Compensation &Performance Appraisal.<br>Marketing Management: Concept, Functions of marketing; Marketing Mix - Product: Stages of Product life cycle, Price, Place: Channels of Distribution&Promotion: Promotion Mix |    |    |   |     |     |        |
| UNIT-III                                                                     |                                                                                                                        | Financial Management: Capital - Need &Significance of capital, Types of Capital - Fixed capital & Working Capital - Components of Working Capital, Factors influencing Working                                                                                                                                                                                                                                                                          |    |    |   |     |     |        |

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| <b>(10Hrs)</b>              | capital. Methods of Raising Finance - Long term, medium term, & Short-term financial sources.<br><b>Strategic Management:</b> Vision, Mission, Goal, Objective, Policy, Strategy. Elements of Corporate planning process; Environmental scanning; SWOT analysis                                                                                                                                                                     |
| <b>UNIT-IV<br/>(10 Hrs)</b> | <b>Organizational Behavior:</b> Concept & features of Organizational Behavior, Individual Behavior Determinants: Perception-Perceptual process; Attitude-Attitudinal change, Personality.<br><b>Motivation:</b> Meaning, Theories of Motivation - Maslow's Theory of Human Needs, Douglas McGregor's Theory X and Theory Y, Herzberg's Two-Factor Theory of Motivation.                                                             |
| <b>UNIT-V<br/>(10 Hrs)</b>  | <b>Managerial Ethics:</b> Ethics and Business, Ethics of Marketing & advertising, Ethics of Finance & Accounting, Decision – making frameworks, Business and Social Responsibility, International Standards, Corporate Governance, Corporate Citizenship, Corporate Social Responsibility<br><b>Stress</b> - Causes & Consequences of stress, Types of stress, Strategies to overcome Stress at Individual and Organisational level |
| <b>Text Books:</b>          |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.                          | Richard L. Daft, Understanding the Theory and Design of Organizations                                                                                                                                                                                                                                                                                                                                                               |
| <b>Reference Books:</b>     |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 1.                          | Stephen P. Robbins, Timothy A. Judge, Niharika Vohra, Organizational Behavior                                                                                                                                                                                                                                                                                                                                                       |

| Code                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Category                                                                                                                                     | L | T  | P  | C | I.M | E.M | Exam   |
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| B20HS2203                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | HS                                                                                                                                           | 3 | -- | -- | 3 | 30  | 70  | 3 Hrs. |
| BUSINESS COMMUNICATION & VALUE SCIENCE-III                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                              |   |    |    |   |     |     |        |
| (For CSBS)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                              |   |    |    |   |     |     |        |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Leadership Oriented Learning (LOL)                                                                                                           |   |    |    |   |     |     |        |
| Nature of Course                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Behavioral                                                                                                                                   |   |    |    |   |     |     |        |
| Pre requisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Basic Knowledge of English (verbal and written) Completion of all units from Semesters 1 and 2                                               |   |    |    |   |     |     |        |
| Course Objectives:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                              |   |    |    |   |     |     |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Develop technical writing skills                                                                                                             |   |    |    |   |     |     |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Introduce students to Self-analysis techniques like SWOT & TOWS                                                                              |   |    |    |   |     |     |        |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Introduce students to key concepts of:<br>a) Pluralism & cultural spaces<br>b) Cross-cultural communication<br>c) Science of Nation building |   |    |    |   |     |     |        |
| Course Outcomes: Upon completion of the course, students shall have ability to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                              |   |    |    |   |     |     |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Apply & analyze the basic principles of SWOT & understand Motivation in life scenarios.                                                      |   |    |    |   |     |     | K3     |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Identify, understand & apply pluralism & cultural communication in Global, glocal and translocational set-ups.                               |   |    |    |   |     |     | K4     |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Applying the scientific feature of grammar and vocabulary in everyday's communication                                                        |   |    |    |   |     |     | K2     |
| 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Understand, apply & analyze the tools of technical writing                                                                                   |   |    |    |   |     |     | K3     |
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Understand the role of science & recognize its impact in daily life                                                                          |   |    |    |   |     |     | K3     |
| Course Contents:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                              |   |    |    |   |     |     |        |
| Objectives for Semester 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                              |   |    |    |   |     |     |        |
| After completing this semester, learners will be able to:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                              |   |    |    |   |     |     |        |
| <ul style="list-style-type: none"><li>Summarize the basic principles of SWOT and Life Positions.</li><li>Apply SWOT in real life scenarios.</li><li>Recognize how motivation helps real life.</li><li>Leverage motivation in real-life scenarios.</li><li>Identify pluralism in cultural spaces.</li><li>Respect pluralism in cultural spaces.</li><li>Differentiate between the different cultures of India.</li><li>Define the terms global, glocal and translocational.</li><li>Differentiate between global, glocal and translocational culture.</li><li>Recognize the implications of cross-cultural communication.</li><li>Identify the common mistakes made in cross-cultural communication.</li><li>Apply cross-cultural communication.</li><li>Differentiate between the roles and relations of different genders.</li><li>Summarize the role of science in nation building.</li><li>Define AI (artificial intelligence).</li><li>Recognize the importance of AI.</li><li>Identify the best practices of technical writing.</li><li>Apply technical writing in real-life scenarios.</li></ul> |                                                                                                                                              |   |    |    |   |     |     |        |

| SYLLABUS                          |                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>UNIT-I</b><br><b>(12 hrs)</b>  | <b>SWOT Vs. TOWS:</b> SWOT and Life Positions, SWOT vs. TOWS, Presentation on SWOT. The Balancing Act (Self Analysis) - Basic principles of SWOT & life positions. Ted talks on biomimicry<br><b>Motivational Skills:</b> Learning motivational skills through Stories and Speeches<br>Maslow's theory - Recognize how motivation helps real life - Leverage motivation in real-life scenarios |
| <b>UNIT-II</b><br><b>(12 hrs)</b> | <b>Pluralism in cultural spaces:</b> Identifying Pluralism and Cultural Spaces. Awareness and respect for pluralism in cultural spaces Theory/Discussion. Global, Glocal and translocational cross-cultural communication: Understanding Global, Glocal, Translocational impacts<br>Cross-Cultural Communication, Culture Shock and Gender Awareness                                           |
| <b>UNIT-III</b><br><b>(8 hrs)</b> | <b>English Grammar &amp; Vocabulary:</b> Error Detection, Voice (Active & passive) Text Completion (Closed/ open)<br>Group Discussion                                                                                                                                                                                                                                                          |
| <b>UNIT-IV</b><br><b>(8 hrs)</b>  | <b>Technical Writing:</b><br>Basic rules of Technical Writing<br>Basic rules of Report Writing and Ad-Making<br>Scenario-based Assessment on technical writing                                                                                                                                                                                                                                 |
| <b>UNIT-V</b><br><b>(8 hrs)</b>   | <b>Role of science in nation building:</b><br>Introduction to Role of science in nation building- Discussion through Augmented Reality, Role of science post- independence<br><b>Presentations:</b> Artificial Intelligence in everyday life<br>"Voice of Future"                                                                                                                              |
| <b>Text Books:</b>                |                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.                                | There are no prescribed textbooks for Semester 4 – there will be handouts and reference links shared.                                                                                                                                                                                                                                                                                          |
| <b>Reference Books:</b>           |                                                                                                                                                                                                                                                                                                                                                                                                |
| 1                                 | Pringle, A. S., & O'Keefe, S. S. (2009). Technical Writing 101: A Real-World Guide to Planning and Writing Technical Content (3rd ed.). Scriptorium Publishing Services, Inc.                                                                                                                                                                                                                  |
| 2                                 | Reynolds, S., Valentine, D., & Munter, M. M. (2019). Guide to Cross-Cultural Communications (2nd Edition) (Guide to Series in Business Communication) (2nd ed.). Pearson.                                                                                                                                                                                                                      |
| 3                                 | Hurn, B., & Tomalin, B. (2016). Cross-Cultural Communication: Theory and Practice (1st ed. 2013 ed.). Palgrave Macmillan.                                                                                                                                                                                                                                                                      |
| 4                                 | Eastwood, John. Oxford Learner's Grammar: Grammar Finder & Checker, 2009. OUP                                                                                                                                                                                                                                                                                                                  |
| 5                                 | Mc Carthy, Michael, & O'Dell, Felicity, English Vocabulary in Use, Second Ed., 2008. CUP                                                                                                                                                                                                                                                                                                       |
| <b>Web References:</b>            |                                                                                                                                                                                                                                                                                                                                                                                                |
| 1                                 | Examples of Technical Writing for Students<br><a href="https://freelance-writing.lovetoknow.com/kinds-technical-writing">https://freelance-writing.lovetoknow.com/kinds-technical-writing</a>                                                                                                                                                                                                  |
| 2                                 | 11 Skills of a Good Technical Writer<br><a href="https://clickhelp.com/clickhelp-technical-writing-blog/11-skills-of-a-good-technical-writer/">https://clickhelp.com/clickhelp-technical-writing-blog/11-skills-of-a-good-technical-writer/</a>                                                                                                                                                |
| 3                                 | 13 benefits and challenges of cultural diversity in the workplace<br><a href="https://www.hult.edu/blog/benefits-challenges-cultural-diversity-workplace/">https://www.hult.edu/blog/benefits-challenges-cultural-diversity-workplace/</a>                                                                                                                                                     |

| <b>Online Resources:</b> |                                                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1                        | <a href="https://youtu.be/CsaTslhSDI">https://youtu.be/CsaTslhSDI</a>                                                                       |
| 2                        | <a href="https://m.youtube.com/watch?feature=youtu.be&amp;v=IIKvV8_T95M">https://m.youtube.com/watch?feature=youtu.be&amp;v=IIKvV8_T95M</a> |
| 3                        | <a href="https://m.youtube.com/watch?feature=youtu.be&amp;v=e80BbX05D7Y">https://m.youtube.com/watch?feature=youtu.be&amp;v=e80BbX05D7Y</a> |
| 4                        | <a href="https://m.youtube.com/watch?v=dT_D68RJ5T8&amp;feature=youtu.be">https://m.youtube.com/watch?v=dT_D68RJ5T8&amp;feature=youtu.be</a> |
| 5                        | <a href="https://m.youtube.com/watch?v=7sLLEdBgYYY&amp;feature=youtu.be">https://m.youtube.com/watch?v=7sLLEdBgYYY&amp;feature=youtu.be</a> |

| Code                                                                             | Category                                                                                                                                                                                                                                                                                                                                                                                                                                    | L  | T  | P | C   | I.M | E.M | Exam   |    |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|--------|----|
| B20CB2204                                                                        | PC                                                                                                                                                                                                                                                                                                                                                                                                                                          | -- | -- | 3 | 1.5 | 15  | 35  | 3 Hrs. |    |
| OS & UNIX PROGRAMMING LAB                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| (For CSBS)                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| Course Objectives: The student who successfully completes this course will have: |                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 1                                                                                | To understand the design aspects of operating system                                                                                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |     |        |    |
| 2                                                                                | To study the process management concepts & Techniques                                                                                                                                                                                                                                                                                                                                                                                       |    |    |   |     |     |     |        |    |
| 3                                                                                | To study the storage management concepts                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        |    |
| 4                                                                                | To familiarize students with the Linux environment                                                                                                                                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |        |    |
| 5                                                                                | To learn the fundamentals of shell scripting/programming                                                                                                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        |    |
| Course Outcomes: After completion of the course, the student will be able to     |                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| S. No                                                                            | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        | KL |
| 1                                                                                | use Unix utilities and perform basic shell control of the utilities                                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |     |        | K2 |
| 2                                                                                | use the Unix file system and file access control                                                                                                                                                                                                                                                                                                                                                                                            |    |    |   |     |     |     |        | K2 |
| 3                                                                                | Implement operating system concepts to develop software                                                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        | K3 |
| 4                                                                                | Solve problems using bash for shell scripting                                                                                                                                                                                                                                                                                                                                                                                               |    |    |   |     |     |     |        | K4 |
| SYLLABUS                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 1                                                                                | a) Study of Unix/Linux general purpose utility command list: man, who, cat, cd, cp, ps, ls, mv, rm, mkdir, rmdir, echo, more, date, time, kill, history, chmod, chown, finger, pwd, cal, logout, shutdown<br>b) Study of vi editor<br>c) Study of Bash shell, Bourne shell and C shell in Unix/Linux operating system<br>d) Study of Unix/Linux file system (tree structure)<br>e) Study of .bashrc, /etc/bashrc and Environment variables. |    |    |   |     |     |     |        |    |
| 2                                                                                | Write a C program that makes a copy of a file using standard I/O, and system calls                                                                                                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |        |    |
| 3                                                                                | Write a C program to emulate the UNIX ls -l command.                                                                                                                                                                                                                                                                                                                                                                                        |    |    |   |     |     |     |        |    |
| 4                                                                                | Write a C program that illustrates how to execute two commands concurrently with a command pipe. Ex: - ls -l   sort                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |     |        |    |
| 5                                                                                | Simulate the following CPU scheduling algorithms: (a) Round Robin (b) SJF (c) FCFS (d) Priority                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 6                                                                                | Multiprogramming-Memory management-Implementation of fork(), wait(), exec() and exit() System calls                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |     |        |    |
| 7                                                                                | Simulate the following: (a) Multiprogramming with a fixed number of tasks (MFT) (b). Multiprogramming with a variable number of tasks (MVT)                                                                                                                                                                                                                                                                                                 |    |    |   |     |     |     |        |    |
| 8                                                                                | Simulate Bankers Algorithm for Dead Lock Avoidance                                                                                                                                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |        |    |
| 9                                                                                | 9. Simulate Bankers Algorithm for Dead Lock Prevention                                                                                                                                                                                                                                                                                                                                                                                      |    |    |   |     |     |     |        |    |
| 10                                                                               | Simulate the following page replacement algorithms: (a)FIFO (b) LRU (c) LFU                                                                                                                                                                                                                                                                                                                                                                 |    |    |   |     |     |     |        |    |
| 11                                                                               | Simulate the following File allocation strategies: (a) Sequenced (b) Indexed (c) Linked                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 12                                                                               | Write a C program that illustrates two processes communicating using shared memory.                                                                                                                                                                                                                                                                                                                                                         |    |    |   |     |     |     |        |    |

| <b>Reference Books:</b> |                                                                    |
|-------------------------|--------------------------------------------------------------------|
| 1.                      | Forouzan and Gilberg, Unix and Shell Programming Cengage Learning. |
| 2.                      | Yashavant and P.Kanetkar, Unix Shell Programming BPB Publishing.   |

| Code                                                                             | Category                                                                                                                                                                                                                                                                                                                                                                                                                                          | L  | T  | P | C   | I.M | E.M | Exam   |    |
|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|--------|----|
| B20CB2205                                                                        | PC                                                                                                                                                                                                                                                                                                                                                                                                                                                | -- | -- | 3 | 1.5 | 15  | 35  | 3 Hrs. |    |
| PYTHON PROGRAMMING LAB                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| (For CSBS)                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| Course Objectives: The student who successfully completes this course will have: |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| 1                                                                                | To acquire programming skills in core Python.                                                                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 2                                                                                | To acquire Object Oriented Skills in Python                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |   |     |     |     |        |    |
| 3                                                                                | To develop the skill of designing Graphical user Interfaces in Python                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        |    |
| 4                                                                                | To develop the ability to write database applications in Python                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| Course Outcomes: After completion of the course, the student will be able to     |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| S. No                                                                            | Outcome                                                                                                                                                                                                                                                                                                                                                                                                                                           |    |    |   |     |     |     |        | KL |
| 1                                                                                | Write, Test and Debug Python Programs                                                                                                                                                                                                                                                                                                                                                                                                             |    |    |   |     |     |     |        | K4 |
| 2                                                                                | Use Conditionals and Loops for Python Programs                                                                                                                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        | K3 |
| 3                                                                                | Use functions and represent Compound data using Lists, Tuples and Dictionaries                                                                                                                                                                                                                                                                                                                                                                    |    |    |   |     |     |     |        | K3 |
| 4                                                                                | Use various user define applications using python                                                                                                                                                                                                                                                                                                                                                                                                 |    |    |   |     |     |     |        | K3 |
| SYLLABUS                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |    |    |   |     |     |     |        |    |
| 1                                                                                | Write a program that asks the user to enter three numbers (use three separate input statements). Create variables called total and average that hold the sum and average of the three numbers and print out the values of total and average.                                                                                                                                                                                                      |    |    |   |     |     |     |        |    |
| 2                                                                                | Write a program that uses a <i>for</i> loop to print the numbers 8, 11, 14, 17, 20, ..., 83, 86, 89.                                                                                                                                                                                                                                                                                                                                              |    |    |   |     |     |     |        |    |
| 3                                                                                | Write a program that asks the user for their name and how many times to print it. The program should print out the user’s name the specified number of times.                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        |    |
| 4                                                                                | Use a <i>for</i> loop to print a triangle like the one below. Allow the user to specify how high the triangle should be.<br>*<br>**<br>***<br>****                                                                                                                                                                                                                                                                                                |    |    |   |     |     |     |        |    |
| 5                                                                                | Write a program that asks the user to enter a word and prints out whether that word contains any vowels.                                                                                                                                                                                                                                                                                                                                          |    |    |   |     |     |     |        |    |
| 6                                                                                | Write a program that asks the user to enter two strings of the same length. The program should then check to see if the strings are of the same length. If they are not, the program should print an appropriate message and exit. If they are of the same length, the program should alternate the characters of the two strings. For example, if the user enters <i>abcde</i> and <i>ABCDE</i> the program should print out <i>AaBbCcDdEe</i> . |    |    |   |     |     |     |        |    |
| 7                                                                                | Write a program that asks the user for a large integer and inserts commas into it according to the standard American convention for commas in large numbers. For instance, if the user enters 1000000, the output should be 1,000,000.                                                                                                                                                                                                            |    |    |   |     |     |     |        |    |
| 8                                                                                | Write a program that generates a list of 20 random numbers between 1 and 100.                                                                                                                                                                                                                                                                                                                                                                     |    |    |   |     |     |     |        |    |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | (a) Print the list.<br>(b) Print the average of the elements in the list.<br>(c) Print the largest and smallest values in the list.<br>(d) Print the second largest and second smallest entries in the list<br>(e) Print how many even numbers are in the list.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 9                       | Write a function called <i>sum_digits</i> that is given an integer num and returns the sum of the digits of num.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 10                      | Write a function called <i>number_of_factors</i> that takes an integer and returns how many factors the number has.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11                      | Write a function called <i>primes</i> that is given a number n and returns a list of the first n primes. Let the default value of n be 100.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12                      | Write a function called <i>merge</i> that takes two already sorted lists of possibly different lengths, and merges them into a single sorted list.<br>(a) Do this using the sort method. (b) Do this without using the sort method.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 13                      | Write a program that asks the user for a word and finds all the smaller words that can be made from the letters of that word. The number of occurrences of a letter in a smaller word can't exceed the number of occurrences of the letter in the user's word.                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 14                      | Write a class called <i>Product</i> . The class should have fields called name, amount, and price, holding the product's name, the number of items of that product in stock, and the regular price of the product. There should be a method <i>get_price</i> that receives the number of items to be bought and returns a the cost of buying that many items, where the regular price is charged for orders of less than 10 items, a 10% discount is applied for orders of between 10 and 99 items, and a 20% discount is applied for orders of 100 or more items. There should also be a method called <i>make_purchase</i> that receives the number of items to be bought and decreases amount by that much. |
| 15                      | Write a class called <i>Time</i> whose only field is a time in seconds. It should have a method called <i>convert_to_minutes</i> that returns a string of minutes and seconds formatted as in the following example: if seconds is 230, the method should return '5:50'. It should also have a method called <i>convert_to_hours</i> that returns a string of hours, minutes, and seconds formatted analogously to the previous method.                                                                                                                                                                                                                                                                        |
| 16                      | Write a Python class to implement <i>pow(x, n)</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 17                      | Write a Python class to reverse a string word by word.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 18                      | Write a program that opens a file dialog that allows you to select a text file. The program then displays the contents of the file in a textbox.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 19                      | Write a program to demonstrate Try/except/else.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 20                      | Write a program to demonstrate try/finally and with/as.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reference Books:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.                      | Introduction to Python Programming, Gowrishankar.S, Veena A, CRC Press.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2.                      | Programming and Problem Solving with Python, Ashok Namdev Kamthane, Amit Ashok Kamthane, TMH, 2019.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

| Code                                                                          | Category                                                                                                                                                                                      | L  | T  | P | C   | I.M | E.M | Exam            |
|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|-----|-----|-----|-----------------|
| B20CB2206                                                                     | PC                                                                                                                                                                                            | -- | -- | 3 | 1.5 | 15  | 35  | 3 Hrs.          |
| DESIGN AND ANALYSIS OF ALGORITHMS LAB                                         |                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| Course Objectives:                                                            |                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| 1                                                                             | Solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking, and branch and bound and writing programs for these solutions |    |    |   |     |     |     |                 |
| 2                                                                             | Analyze the asymptotic performance of algorithms.                                                                                                                                             |    |    |   |     |     |     |                 |
| 3                                                                             | Demonstrate a familiarity with major algorithms and data structures                                                                                                                           |    |    |   |     |     |     |                 |
| 4                                                                             | Synthesize efficient algorithms in common engineering design situations.                                                                                                                      |    |    |   |     |     |     |                 |
| Course Outcomes: Upon Completion of the course, the students will be able to: |                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| S. No                                                                         | Outcome                                                                                                                                                                                       |    |    |   |     |     |     | Knowledge Level |
| 1                                                                             | Apply the Divide-and-Conquer strategy and Greedy Method for solving the complex problems & analyze the performance of solutions.                                                              |    |    |   |     |     |     | K3              |
| 2                                                                             | Apply the optimistic strategies Dynamic Programming for computational problems in computer field.                                                                                             |    |    |   |     |     |     | K3              |
| 3                                                                             | Apply the Backtracking and Branch-and-bound strategies for solving complex problems                                                                                                           |    |    |   |     |     |     | K3              |
| 4                                                                             | Design and implement different Graph algorithms to solve different problems                                                                                                                   |    |    |   |     |     |     | K4              |
| LIST OF EXPERIMENTS                                                           |                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| 1                                                                             | Write a program to implement Quick sort algorithm for sorting a list of integers in ascending order                                                                                           |    |    |   |     |     |     |                 |
| 2                                                                             | Write a program to implement Merge sort algorithm for sorting a list of integers in ascending order.                                                                                          |    |    |   |     |     |     |                 |
| 3                                                                             | Write a program to implement the DFS algorithm for a graph.                                                                                                                                   |    |    |   |     |     |     |                 |
| 4                                                                             | Write a. program to implement the BFS algorithm for a graph.                                                                                                                                  |    |    |   |     |     |     |                 |
| 5                                                                             | Write a program to implement backtracking algorithm for the N-queens problem.                                                                                                                 |    |    |   |     |     |     |                 |
| 6                                                                             | Write a program to implement the backtracking algorithm for the sum of subsets problem.                                                                                                       |    |    |   |     |     |     |                 |
| 7                                                                             | Write a program to implement the backtracking algorithm for the Hamiltonian Circuits problem.                                                                                                 |    |    |   |     |     |     |                 |
| 8                                                                             | Write a program to implement greedy algorithm for job sequencing with deadlines.                                                                                                              |    |    |   |     |     |     |                 |
| 9                                                                             | Write a program to implement Dijkstra’s algorithm for the Single source shortest path problem.                                                                                                |    |    |   |     |     |     |                 |
| 10                                                                            | Write a program that implements Prim’s algorithm to generate minimum cost spanning tree.                                                                                                      |    |    |   |     |     |     |                 |
| 11                                                                            | Write a program that implements Kruskal’s algorithm to generate minimum cost spanning tree                                                                                                    |    |    |   |     |     |     |                 |
| 12                                                                            | Write a program to implement Floyd’s algorithm for the all pairs shortest path problem.                                                                                                       |    |    |   |     |     |     |                 |
| 13                                                                            | Write a program to implement Dynamic Programming algorithm for the 0/1 Knapsack problem.                                                                                                      |    |    |   |     |     |     |                 |
| 14                                                                            | Write a program to implement Dynamic Programming algorithm for the Optimal Binary Search Tree Problem.                                                                                        |    |    |   |     |     |     |                 |
| Reference Books:                                                              |                                                                                                                                                                                               |    |    |   |     |     |     |                 |
| 1.                                                                            | Ellis Horowitz, SartajSahni, Sanguthevar Rajasekaran, “Fundamentals of Computer Algorithms”, 2nd Edition, Universities Press.                                                                 |    |    |   |     |     |     |                 |
| 2.                                                                            | Harsh Bhasin, “Algorithms Design & Analysis”, Oxford University Press.                                                                                                                        |    |    |   |     |     |     |                 |
|                                                                               |                                                                                                                                                                                               |    |    |   |     |     |     |                 |

| Code                                                                       | Category                                                                                                                                                                                                                              | L | T  | P | C | I.M | E.M | Exam  |
|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|---|-----|-----|-------|
| B20IT2207                                                                  | SOC                                                                                                                                                                                                                                   | - | -- | 4 | 2 | --  | 50  | 3 Hrs |
| ANIMATIONS                                                                 |                                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
| (Skill Oriented Course-II)                                                 |                                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
| (Common to AIDS, CSBS & IT)                                                |                                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
| Course Objectives: The objectives of the course are to impart:             |                                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
| 1.                                                                         | This Course will enable students to learn various aspects of animation using a variety of 2-D software and to implement advance principles of traditional animation in Adobe animate to create high quality animation for production. |   |    |   |   |     |     |       |
| Course outcomes : After completion of the course, students will be able to |                                                                                                                                                                                                                                       |   |    |   |   |     |     | K L   |
| 1                                                                          | Learn various tools of digital 2-D animation.                                                                                                                                                                                         |   |    |   |   |     |     | K2    |
| 2                                                                          | Understand production pipeline to create 2-D animation.                                                                                                                                                                               |   |    |   |   |     |     | K2    |
| 3                                                                          | Analyze special effects in animation to bring interest and awe in the scenes and Back grounds.                                                                                                                                        |   |    |   |   |     |     | K3    |
| 4                                                                          | Apply the tools to create 2D animation for films and videos.                                                                                                                                                                          |   |    |   |   |     |     | K3    |
| SYLLABUS                                                                   |                                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
| Adobe Photoshop                                                            | Create your visiting card                                                                                                                                                                                                             |   |    |   |   |     |     |       |
|                                                                            | Create Title for any forthcoming film                                                                                                                                                                                                 |   |    |   |   |     |     |       |
|                                                                            | Digital Matte Paint                                                                                                                                                                                                                   |   |    |   |   |     |     |       |
|                                                                            | Convert Black and White to Color                                                                                                                                                                                                      |   |    |   |   |     |     |       |
|                                                                            | Convert Day mode to Night mode                                                                                                                                                                                                        |   |    |   |   |     |     |       |
|                                                                            | Design Image manipulation                                                                                                                                                                                                             |   |    |   |   |     |     |       |
|                                                                            | Smooth skin and remove blemishes & scars                                                                                                                                                                                              |   |    |   |   |     |     |       |
|                                                                            | Create a 3D pop-out effect                                                                                                                                                                                                            |   |    |   |   |     |     |       |
|                                                                            | Create Textures                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
|                                                                            | Timeline Animation                                                                                                                                                                                                                    |   |    |   |   |     |     |       |
| Adobe Illustrator                                                          | Advertisement                                                                                                                                                                                                                         |   |    |   |   |     |     |       |
|                                                                            | Digital Illustrations                                                                                                                                                                                                                 |   |    |   |   |     |     |       |
|                                                                            | Brochure                                                                                                                                                                                                                              |   |    |   |   |     |     |       |
|                                                                            | Packet Design(Toothpaste packet, Soap cover, any Food product)                                                                                                                                                                        |   |    |   |   |     |     |       |
|                                                                            | Danglers for display                                                                                                                                                                                                                  |   |    |   |   |     |     |       |
|                                                                            | Menu cards                                                                                                                                                                                                                            |   |    |   |   |     |     |       |
|                                                                            | Calendar Design                                                                                                                                                                                                                       |   |    |   |   |     |     |       |
|                                                                            | Tracing image                                                                                                                                                                                                                         |   |    |   |   |     |     |       |
|                                                                            | Vehicle Design                                                                                                                                                                                                                        |   |    |   |   |     |     |       |
|                                                                            | Festival                                                                                                                                                                                                                              |   |    |   |   |     |     |       |
| Adobe In design                                                            | Magazine A4 Size                                                                                                                                                                                                                      |   |    |   |   |     |     |       |
|                                                                            | Newspaper layout design & advertisements – Fine arts                                                                                                                                                                                  |   |    |   |   |     |     |       |

|                         |                                                                        |
|-------------------------|------------------------------------------------------------------------|
|                         | Special Supplement                                                     |
|                         | Different categories of Books                                          |
|                         | Info-graphics                                                          |
|                         | Caricatures                                                            |
|                         |                                                                        |
| <b>Corel<br/>DRAW</b>   | Create a paper ad for advertising of any commercial agency             |
|                         | Package Design                                                         |
|                         | Corporate ID                                                           |
|                         | Exhibition Layout                                                      |
|                         | Oblers                                                                 |
|                         |                                                                        |
| <b>Animation</b>        | Creating Web Banners in Adobe Flash                                    |
|                         | Creating a Logo Animation in Adobe Flash                               |
|                         | Creating Frame by Frame animation                                      |
|                         | Draw Cartoon Animation using reference.                                |
|                         | Create Lip Sink to Characters                                          |
|                         | Using filters & Special effects                                        |
|                         | Create a scene by using Mask layers animation                          |
|                         | <b>E-Learning Lab:</b>                                                 |
|                         | Student Application form                                               |
|                         | Video Controlling                                                      |
|                         | Audio Controlling                                                      |
|                         | Start Drag and Stop Drag Actions                                       |
|                         | Interactive Keyboard Controls using Flash Action Script.               |
|                         | Interactive Flash Game.                                                |
|                         | Creating Character Animation in After Effects                          |
|                         |                                                                        |
| <b>Reference Books:</b> |                                                                        |
| 1                       | Adobe Animate CC Classroom Book 2018 Animation, First Edition, Pearson |
|                         |                                                                        |

| Code                                                                       | Category                                                                   | L | T  | P | C | I.M | E.M | Exam  |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------|---|----|---|---|-----|-----|-------|
| B20IT2208                                                                  | SOC                                                                        | - | -- | 4 | 2 | --  | 50  | 3 Hrs |
|                                                                            |                                                                            |   |    |   |   |     |     |       |
| WEB PAGE DESIGN USING PHP                                                  |                                                                            |   |    |   |   |     |     |       |
| (Skill Oriented Course-II)                                                 |                                                                            |   |    |   |   |     |     |       |
| (Common to AIDS, CSBS & IT)                                                |                                                                            |   |    |   |   |     |     |       |
| Course Objectives: The objectives of the course are to impart:             |                                                                            |   |    |   |   |     |     |       |
| 1.                                                                         | Understand the principles of creating an effective web page.               |   |    |   |   |     |     |       |
| 2.                                                                         | Understand elements of design with regard to the web.                      |   |    |   |   |     |     |       |
| 3.                                                                         | Learn the language of the web: HTML and CSS.                               |   |    |   |   |     |     |       |
| 4.                                                                         | Develop skills in analyzing the usability of a web site.                   |   |    |   |   |     |     |       |
| 5.                                                                         | Understand how to develop PHP web application with database connectivity.  |   |    |   |   |     |     |       |
| 6.                                                                         | Learn CSS grid layout.                                                     |   |    |   |   |     |     |       |
|                                                                            |                                                                            |   |    |   |   |     |     |       |
| Course outcomes : After completion of the course, students will be able to |                                                                            |   |    |   |   |     |     | K L   |
| 1                                                                          | Apply the principles of creating an effective web page.                    |   |    |   |   |     |     | K3    |
| 2                                                                          | Apply the elements of design with regard to the web.                       |   |    |   |   |     |     | K3    |
| 3                                                                          | Create the language of the web: HTML and CSS.                              |   |    |   |   |     |     | K4    |
| 4                                                                          | Develop skills in analyzing the usability of a web site.                   |   |    |   |   |     |     | K4    |
| 5                                                                          | Understand how to plan and conduct user research related to web usability. |   |    |   |   |     |     | K2    |
| 6                                                                          | Create CSS grid layout                                                     |   |    |   |   |     |     | K4    |
| SYLLABUS                                                                   |                                                                            |   |    |   |   |     |     |       |
| Exercise 1                                                                 | Introduction to HTML                                                       |   |    |   |   |     |     |       |
|                                                                            | 1.1 What is HTML                                                           |   |    |   |   |     |     |       |
|                                                                            | 1.2 HTML Documents                                                         |   |    |   |   |     |     |       |
|                                                                            | 1.3 Basic structure of an HTML document                                    |   |    |   |   |     |     |       |
|                                                                            | 1.4 Creating an HTML document                                              |   |    |   |   |     |     |       |
|                                                                            | 1.5 Mark up Tags                                                           |   |    |   |   |     |     |       |
|                                                                            | 1.6 Heading-Paragraphs                                                     |   |    |   |   |     |     |       |
|                                                                            | 1.7 Line Breaks                                                            |   |    |   |   |     |     |       |
|                                                                            | 1.8 HTML Tags.                                                             |   |    |   |   |     |     |       |
|                                                                            |                                                                            |   |    |   |   |     |     |       |
| Exercise 2                                                                 | Elements of HTML                                                           |   |    |   |   |     |     |       |
|                                                                            | 2.1 Introduction to elements of HTML                                       |   |    |   |   |     |     |       |
|                                                                            | 2.2 Working with Text                                                      |   |    |   |   |     |     |       |
|                                                                            | 2.3 Working with Lists, Tables and Frames                                  |   |    |   |   |     |     |       |
|                                                                            | 2.4 Working with Hyperlinks, Images and Multimedia                         |   |    |   |   |     |     |       |
|                                                                            | 2.5 Working with Forms and controls.                                       |   |    |   |   |     |     |       |
|                                                                            |                                                                            |   |    |   |   |     |     |       |
| Exercise 3                                                                 | Introduction to Cascading Style Sheets                                     |   |    |   |   |     |     |       |
|                                                                            | 3.1 Concept of CSS                                                         |   |    |   |   |     |     |       |
|                                                                            | 3.2 Creating Style Sheet                                                   |   |    |   |   |     |     |       |
|                                                                            | 3.3 CSS Properties                                                         |   |    |   |   |     |     |       |
|                                                                            | 3.4 CSS Styling(Background, Text Format, Controlling Fonts)                |   |    |   |   |     |     |       |
|                                                                            | 3.5 Working with block elements and objects                                |   |    |   |   |     |     |       |

|                         |                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                         | 3.6 Working with Lists and Tables                                                                                                     |
|                         | 3.7 CSS Id and Class                                                                                                                  |
|                         | 3.8 Box Model (Introduction, Border properties, Padding Properties, Margin properties)                                                |
| <b>Exercise 4</b>       | 4.1 The Basic of JavaScript: Objects,                                                                                                 |
|                         | 4.2 Primitives Operations and Expressions,                                                                                            |
|                         | 4.3 Screen Output and Keyboard Input,'                                                                                                |
|                         | 4.4 Object Creation and Modification, Arrays, Functions                                                                               |
|                         | 4.5 DHTML: Positioning Moving and Changing Elements                                                                                   |
|                         |                                                                                                                                       |
| <b>Exercise 5</b>       | 5.1 Introducing PHP: Creating PHP script,                                                                                             |
|                         | 5.2 Running PHP script.                                                                                                               |
|                         | 5.3 Using variables, constants, Datatypes, Operators.                                                                                 |
|                         | 5.4 Conditional statements, Control statements, Arrays ,functions                                                                     |
|                         | 5.5 Working with forms and Databases such as MySQL.                                                                                   |
|                         | 5.6 Develop PHP MySQL CRUD Application                                                                                                |
|                         |                                                                                                                                       |
| <b>Text Books:</b>      |                                                                                                                                       |
| 1.                      | Web Technologies, Uttam K Roy, Oxford                                                                                                 |
| 2.                      | HTML 5 Black Book (Covers CSS3, JavaScript, XML, XHTML, AJAX, PHP, jQuery) 2Ed, Dreamtech Press; Second edition                       |
| 3.                      | The Web Warrior Guide to Web Programming, Bai, Ekedahl, Farrelly, Gosselin, Zak, Karparhi, MacIntyre, Morrissey, Cengage              |
|                         |                                                                                                                                       |
| <b>Reference Books:</b> |                                                                                                                                       |
| 1.                      | Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5, Shroff Publishers & Distributors Private Limited - Mumbai; Fourth edition |
| 2.                      | PHP: The Complete Reference, McGraw Hill Education; Raunak php study edition                                                          |

| Code                                            | Category                                                                                                                            | L | T  | P  | C  | I.M | E.M | Exam            |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---|----|----|----|-----|-----|-----------------|
| B20MC2201                                       | MC                                                                                                                                  | 2 | -- | -- | -- | --  | --  | --              |
|                                                 |                                                                                                                                     |   |    |    |    |     |     |                 |
| ENGLISH PROFICIENCY                             |                                                                                                                                     |   |    |    |    |     |     |                 |
| (Common to CE,EEE,ME,AIDS & CSBS)               |                                                                                                                                     |   |    |    |    |     |     |                 |
| Course Objectives: The students will be able to |                                                                                                                                     |   |    |    |    |     |     |                 |
| 1.                                              | Communicate their ideas and views effectively                                                                                       |   |    |    |    |     |     |                 |
| 2.                                              | Practice language skills and improve their language competency.                                                                     |   |    |    |    |     |     |                 |
| 3.                                              | Know and perform well in real life contexts                                                                                         |   |    |    |    |     |     |                 |
| 4.                                              | Identify and examine their self-attributes which require improvement and motivation.                                                |   |    |    |    |     |     |                 |
| 5.                                              | Build confidence and overcome their inhibitions, stage freight, nervousness etc.,                                                   |   |    |    |    |     |     |                 |
| 6.                                              | Improve their reading skills.                                                                                                       |   |    |    |    |     |     |                 |
|                                                 |                                                                                                                                     |   |    |    |    |     |     |                 |
| Course Outcomes: The students will              |                                                                                                                                     |   |    |    |    |     |     |                 |
| S.No                                            | Outcome                                                                                                                             |   |    |    |    |     |     | Knowledge Level |
| 1.                                              | Improve speaking skills.                                                                                                            |   |    |    |    |     |     | K3              |
| 2.                                              | Enhance their listening capabilities                                                                                                |   |    |    |    |     |     | K3              |
| 3.                                              | Learn and practice the skills of composition writing.                                                                               |   |    |    |    |     |     | K3              |
| 4.                                              | Enhance their reading and understanding of different texts.                                                                         |   |    |    |    |     |     | K3              |
| 5.                                              | Improve their communication both in formal and informal contexts.                                                                   |   |    |    |    |     |     | K3              |
| 6.                                              | Be confident in presentation skills.                                                                                                |   |    |    |    |     |     | K3              |
| SYLLABUS                                        |                                                                                                                                     |   |    |    |    |     |     |                 |
| UNIT-I                                          | Listening Skills<br>Types of listening<br>Hearing and Listening<br>Listening as a receptive skill                                   |   |    |    |    |     |     |                 |
|                                                 |                                                                                                                                     |   |    |    |    |     |     |                 |
| UNIT-II                                         | Speaking Skills<br>Presentation skills<br>Describing event/place/thing<br>Extempore<br>Debate<br>Group Discussion                   |   |    |    |    |     |     |                 |
|                                                 |                                                                                                                                     |   |    |    |    |     |     |                 |
| UNIT-III                                        | Reading Skills<br>Types of Reading (Intensive and Extensive reading, Skimming, Scanning)<br>Reading/Summarizing News Paper Articles |   |    |    |    |     |     |                 |
|                                                 |                                                                                                                                     |   |    |    |    |     |     |                 |
| UNIT-IV                                         | Writing Skills<br>Essay Writing (Argumentative, Analytical and Descriptive)<br>E-Mail Writing<br>Business Letters<br>Resume Writing |   |    |    |    |     |     |                 |

|                                                                   |                                                                                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
|                                                                   |                                                                                                                           |
| <b>UNIT-V</b>                                                     | <b>Integrated Language Skills</b><br>Listening Skills for Speaking and Writing<br>Reading Skills for Writing and Speaking |
|                                                                   |                                                                                                                           |
| <b>Reference Books:</b>                                           |                                                                                                                           |
| 1.                                                                | Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP                                          |
| 2.                                                                | English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009    |
| 3.                                                                | Enriching Speaking and Writing Skills, Orient Blackswan Publishers.                                                       |
| 4.                                                                | The Oxford Guide to Writing and Speaking by John Seely OUP.                                                               |
| 5.                                                                | Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw hill.                                                     |
| 6.                                                                | Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers                                                         |
|                                                                   |                                                                                                                           |
| Note: Internal Assessment is carried out throughout the semester. |                                                                                                                           |